



University of Buenos Aires
Faculty of Exact and Natural Sciences

Temporal and regional analysis of customer satisfaction *drivers*

Licenciatura Thesis in Data Science

Cristian Antonio

Supervisor: Dr. Sebastián Pinto

Co-supervisors: Dr. Caterina Mosto and Lic. Santiago Kahane

Buenos Aires, 2026

TEMPORAL AND REGIONAL ANALYSIS OF CUSTOMER SATISFACTION *DRIVERS*

This thesis conducts a temporal and regional analysis of the determinants (drivers) of customer satisfaction in a supermarket chain between June 2021 and March 2026. Using responses to satisfaction questionnaires, it starts from a simple linear model and progressively incorporates different strategies to address the main challenges of the problem. First, to address the structural fragmentation caused by different versions of the questionnaire and the presence of missing or non-evaluative responses, a variable harmonization process was implemented together with multiple imputation by chained equations (MICE) algorithms. Then, to address the strong imbalance in the overall satisfaction variable, a strategy based on weighted least squares (WLS), combined with sequential feature selection (SFS), was adopted. This made it possible to obtain a reduced model of $K = 20$ interpretable variables without losing generalization ability. The thesis also analyzes potential asymmetries in the effects of the drivers through a model consistent with the logic of loss aversion. The results show that, for some attributes, a poor evaluation can penalize overall satisfaction more than a good evaluation increases it. Finally, the analysis by store format (Large, Medium, and Small) shows that the relevance of the drivers is not entirely homogeneous, and that segmentation makes it possible to identify important differences in how customer satisfaction is formed.

Keywords: Satisfaction drivers, Grocery Retail, Loss Aversion, Sequential Feature Selection.

ACKNOWLEDGEMENTS

It has been a difficult journey. With this thesis, I am bringing to a close not only an academic chapter, but also a chapter of personal growth. I have reached a place I thought I would never reach. None of this would have been possible without my parents, who were always there and supported me through my hardest moments. I owe them everything good in my life. They taught me never to give up. Mom, Dad, Sonia, Carlos: thank you for being my parents and for helping me fight to make my dreams come true.

To my sister, Carla, for being by my side since the beginning of my life and for always being there. For helping me grow and become a better person. And also for giving me, together with Fabi, the joy of being an uncle to the most special little person in the world, Elena.

To my cousin Pablo, who is more a brother than a cousin, and who stood by me through some of my hardest moments. He is also the main reason I chose to study at Exactas.

To my whole family, my cousins, and my aunts and uncles. You all contributed in some way to my being here today, giving me love and support whenever I needed it. To my grandparents and my aunt Tita, who are no longer physically here, but live on in my memories. Many of their words and pieces of advice are still with me and help me keep going.

To my lifelong friends, Tomás, Luciano, and Isidro. A whole lifetime together, guys. Thank you for being part of this journey and for always supporting me. And also to Mati, whom I met seven years ago, but who feels like a lifelong friend.

To my Mathematics classmates, who made my first years at university so much more manageable and filled them with good moments. Without them, I could not have dealt with the frustration of those difficult early years.

To the Data Science guys, including Manu, Nacho, Echi, Tincho, Eze, and Mateo. For helping me reach this goal at different stages of the degree. Taking classes with you was so much fun.

To all the faculty members at the UBA Faculty of Exact and Natural Sciences, for giving me an education I am proud of. To my supervisors, Sebastián, Caterina, and Santiago, for guiding me throughout the process. This thesis would not have been possible without your help. A thousand heartfelt thanks.

I love you all.

Cristian

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INTRODUCTION

Understanding and improving the consumer experience in the supermarket retail industry (grocery retail) is an important problem for understanding how customer satisfaction is formed. In recent decades, competition in this sector has intensified, establishing large retail chains as players that account for a substantial share of the overall market [1]. Unlike other economic sectors, where transactions are infrequent or mediated by long-term contracts, supermarkets are characterized by frequent purchases, direct exposure to multiple physical touchpoints, and virtually no exit barriers or switching costs for the customer. In this competitive context, designing satisfying shopping experiences becomes central to strengthening customer retention and sustaining long-term profitability [2]. This is because, in the food retail industry, the relationship with the consumer depends critically on the experience within the store. During a shopping trip, the customer interacts with many service elements at the same time, which collectively affect perceived overall performance. Consequently, consumer satisfaction is not only a descriptive indicator, but also a useful variable for understanding and managing the experience.

A major change in supermarket retail in recent decades has been the diversification of store formats [3, 4, 5]. On the one hand, hypermarkets are characterized by the convenience of doing all one's shopping in one place [3, 6]. On the other hand, traditional supermarkets, although they generally offer a narrower product assortment, compete by providing more personalized service and an experience tailored to local needs [7]. Convenience stores, in turn, are defined by a limited assortment and quick shopping trips. This classification suggests that consumers visit each format according to their needs [8]. Consequently, product-related attributes, such as variety and prices, become more relevant for larger shopping trips, whereas service and convenience attributes carry greater weight in smaller purchases. Purchase frequency itself [9] can also lead customers to pay more attention to certain attributes than to others: a customer who frequently visits a convenience store will prioritize accessibility and checkout speed, whereas on regular supermarket shopping trips they may balance store proximity, low prices, and continuous stock availability.

To address these dimensions of the in-store shopping experience quantitatively, the literature on retail service quality offers different frameworks. Among them is the Retail Service Quality Scale (RSQS) [10], which breaks down the consumer's evaluation into blocks such as the physical aspects of the infrastructure, reliability in fulfilling commercial promises, interactions with staff, problem solving at the point of sale, and commercial policies. This conceptual structure makes it possible to classify a questionnaire's matrix of questions into blocks of drivers for mathematical modeling. The main reference work for this thesis follows this approach: Goić, Levenier, and Montoya (2021) [11] propose a longitudinal analysis showing that the drivers of behavior and satisfaction do not have a uniform impact across

an entire retail chain; instead, their relative relevance may depend on the store format the consumer chooses for their shopping.

An important aspect of this area of research is questioning the traditional assumption of symmetry in the impact of service attributes on overall satisfaction. Pioneering research in marketing and consumer behavior, consolidated by Anderson and Mittal [12], shows that the relationship between the performance of a specific attribute and overall satisfaction may be asymmetric, challenging traditional assumptions of symmetric linearity. This premise is associated with the foundations of behavioral economics and the psychology of Prospect Theory, formulated by Kahneman and Tversky [13], which posits that individuals may exhibit marked loss aversion, processing negative stimuli more intensely than positive stimuli of equal magnitude.

Despite this empirical evidence, part of the customer satisfaction literature assumes that the functional relationship between individual customers' responses to service and their overall satisfaction can be characterized by a linear function. However, various studies suggest that a nonlinear response function that explicitly incorporates the concept of loss aversion allows the impact of perceived service quality to be modeled with a more flexible interpretation [14, 15]. Under this approach, the effect of operational performance is assessed asymmetrically relative to a reference point determined by the customer's own prior expectations; the response may be more pronounced in the loss region, when service falls below expectations, than in the gain region, when it exceeds the expectation threshold. Applied to supermarket retail, this suggests that certain attributes may serve as minimum standards expected by customers. In other words, good performance does not necessarily produce a proportional improvement in satisfaction, but a poor evaluation may impose a greater penalty on the overall experience.

The objective of this thesis is to conduct a temporal and regional analysis of customer satisfaction drivers in a supermarket chain whose identity is withheld for confidentiality reasons, over the period from June 2021 to March 2026, evaluating the stability of their coefficients and identifying the presence of asymmetric effects consistent with the logic of loss aversion. To achieve this aim, specific objectives are established, first, to develop a harmonization and consolidation process that enables the analysis of the satisfaction surveys. Second, the study seeks to estimate and compare the performance of interpretable linear models with models of discrete indicator variables that separately capture the cost of a negative evaluation and the benefit of a high evaluation. Finally, it evaluates whether segmenting the modeling by store format provides additional information for interpreting and predicting satisfaction.

Under confidentiality agreements with the organization that provided the data, this version of the document omits the company's identity as well as the results chapters. Chapter 1 presents the methodological aspects of survey collection and the process of unifying and preprocessing the raw data. Chapter 2 is omitted from this version. Chapter 3 describes the modeling methodology: the simple weighted linear model, dimensionality reduction algorithms using sequential feature selection (SFS), and the categorical indicator variable model designed to test empirically for asymmetry consistent with loss aversion. Chapter 4 is omitted from this version. Finally, the Conclusions chapter synthesizes the main aspects

of the study. Methodological details are provided in the Appendix, and the main libraries used in the computational implementation are presented in Appendix E.

1. DATA AND DATASET CLEANING

This first chapter describes the surveys from which the data used in this thesis were obtained. It then presents the process of cleaning and unifying the original files to obtain the final dataset.

1.1. Surveys

The data come from a supermarket chain whose identity is withheld for confidentiality reasons.

A set of files provided by the company is available for the development of this thesis. These records contain responses obtained through customer satisfaction surveys conducted monthly or every two months during the period from June 2021 to March 2026. Throughout this thesis, each of the time windows (of one or two months) into which the company grouped the collection of satisfaction survey responses will be called a “survey wave.” The surveys were conducted online in a non-transactional format (after the shopping experience, but not immediately afterward), with the aim of evaluating the service provided in the different stores. Participants also had access to an incentive system associated with a points program and could respond only once per wave.

The questionnaire has four versions over time, resulting from minor modifications made to improve it. Figure 1.1 shows how the 34,595 responses are distributed over time. Each bar represents a survey wave, of which there are 40 in total, and its color corresponds to the questionnaire type in use during each period.

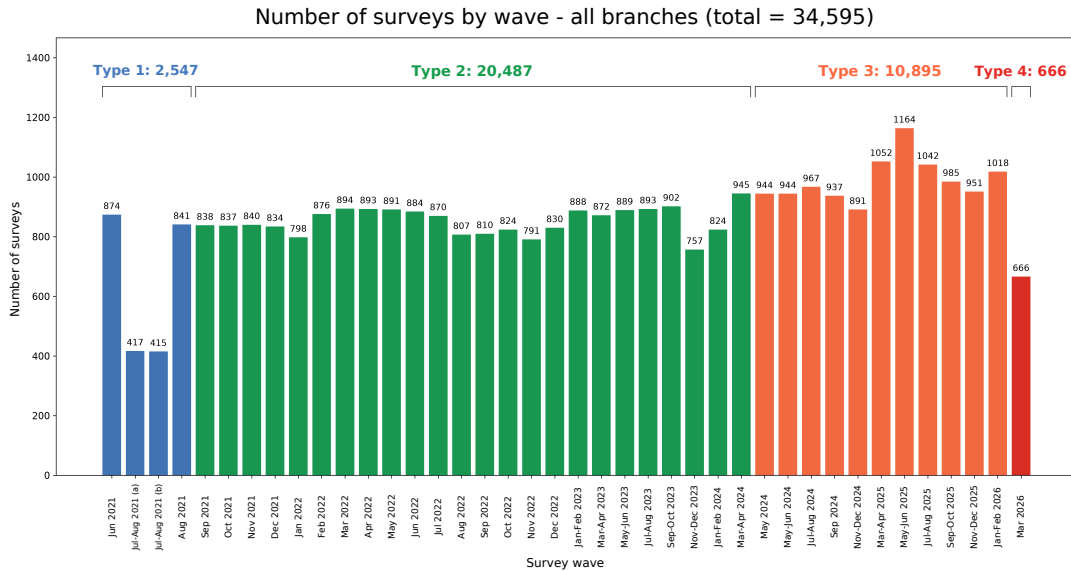


Fig. 1.1: Number of surveys completed by survey wave and questionnaire type.

Although each questionnaire type has its own particular features, a general core is maintained throughout the period. This core includes questions associated with the main

supermarket, the store or shopping channel, overall satisfaction, and different aspects of the in-store experience, such as infrastructure, variety and availability, quality, prices, and service, as well as demographic aspects such as gender and age. Beyond this core, there are questions related to e-commerce, competitors, the Net Promoter Score (NPS, a measure of willingness to recommend the chain or another supermarket), advertising/communication, or other specific sections depending on the version. What constitutes a questionnaire type is analyzed in detail in Appendix A.1.

It is also important to emphasize that, within a single questionnaire, different response paths could exist depending on the characteristics and services offered by the store visited. Table 1.1 summarizes the general structure of the questionnaires by response type. This review identified 10 main blocks, showing that the questionnaire organizes questions around thematic dimensions.

Tab. 1.1: Response types and content of the questionnaire blocks

Block	Response type	Description
General	Nominal categorical, ordinal, and numerical	Includes general classification variables, such as ranking, store, last visit, and overall satisfaction.
Demographics	Nominal and ordinal categorical	Includes respondent characteristics, mainly age and gender.
Infrastructure	Ordinal 1–4 and non-evaluative codes	Evaluates physical characteristics of the store, such as cleanliness, restrooms, shopping carts, baskets, and other aspects of infrastructure.
Assortment/availability	Ordinal 1–4, non-evaluative codes, and, in some cases, associated free text	Evaluates product variety and availability.
Quality	Ordinal 1–4, non-evaluative codes, and, in some cases, associated free text	Mainly evaluates the quality/freshness of products and specific departments.
Value for money (RCP)	Ordinal 1–4, non-evaluative codes, and, in some cases, associated free text	Evaluates the customer’s perception of the relationship between price and quality.
Prices	Ordinal 1–4, non-evaluative codes, and, in some cases, associated free text	Evaluates prices by category, overall prices, and the attractiveness of special offers.
Customer service	Ordinal 1–4, non-evaluative codes, and, in some cases, associated free text	Evaluates aspects related to service from staff, staff availability, and waiting times.
Competitor comparison	Ordinal comparative, NPS, and associated free text	Includes comparisons with the chain and other supermarkets, as well as recommendation questions.
Advertising and campaigns	Categorical, binary or multiple selection, ordinal, and associated free text	Includes questions related to campaigns, advertising, and recall of marketing activities.

Note: On the ordinal 1–4 scales, lower values indicate a more favorable evaluation. In many cases, the value 5 corresponds to non-evaluative responses, such as “I do not use it,” “Do not know,” or equivalent responses. Depending on the question, special codes, such as 4 or 99, may also appear for categories such as “Do not know,” “No answer,” or equivalent responses. Some blocks also include free-text fields associated with evaluative responses.

As shown in Table 1.1, numerical responses predominate in all blocks, with the possibility of accompanying free text in some cases.

1.1.1. Cleaning and unifying the dataset

The structural variation among survey types created the need to construct a unified dataset. This stage was essential because of the structural changes the questionnaires underwent over time: the variables needed to be standardized to consolidate a robust, consistent dataset suitable for subsequent analysis.

First, survey coverage by store over time was analyzed. Only physical stores with complete coverage over time were considered. This criterion left 13 stores. With this restriction, the sample

was reduced to 26,480 observations. In addition, because 2 files (corresponding to 2 survey waves) had missing data in several relevant blocks, they were excluded from the final dataset used for the analysis, reducing the sample to 25,117 observations.

Because the different questionnaire types have structural differences, a unification process was necessary. This consisted of identifying equivalent questions within each questionnaire (associated with different response paths), collapsing those that represented the same aspect being evaluated within a single wave, and then bringing the resulting variables into a common structure. The procedure is described in detail in Appendix A.2.

The “Competitor comparison” and “Advertising/Campaigns” blocks were excluded from the unification process. The “Competitor comparison” block was excluded because these features do not directly explain specific aspects of the customer’s in-store experience, making them difficult to interpret within the model. The “Advertising/Campaigns” block, in turn, showed substantial variation in its questions, both between survey types and between waves within the same type. These questions also have response types that are more difficult to interpret and would therefore require a more extensive analysis.

For each variable retained after unification, its presence in each version of the questionnaire was verified, along with the classification of each question into blocks. These modules are intended to divide the questionnaire into different sections, which may or may not be available at each store depending on the services provided.

Most variables were found to remain present in the questionnaires throughout the period analyzed. As mentioned previously, some are only partially available because of the changes made over time.

Because the target variable modeled in this thesis is overall satisfaction, it is necessary to clarify how it was handled in the initial descriptive analyses. In the survey, overall satisfaction is recorded using ordinal categories: “Completely satisfied,” “Satisfied,” “Slightly satisfied,” and “Dissatisfied.” Because these categories have a clear order, at this first stage they were mapped to an initial numerical scale, where 1 corresponds to “Completely satisfied,” 2 to “Satisfied,” 3 to “Slightly satisfied,” and 4 to “Dissatisfied.”

3. MODELING SATISFACTION

This chapter examines how evaluations of different aspects of the in-store experience relate to the overall satisfaction reported by respondents. To do so, the analysis began with a simple linear model and then incorporated various modifications as problems specific to the sample emerged. In particular, the treatment of problematic cells, the imbalance in the response variable, and the number of available drivers were considered.

After exploring several alternatives, a main model was selected to interpret the drivers associated with overall satisfaction. The aim was to retain a clear interpretation of the coefficients without relying on an excessively unstable fit.

For confidentiality reasons, this version presents only the modeling methodology: the estimated coefficients, the drivers identified, and their interpretation are omitted.

3.1. A simple satisfaction model

This section seeks to estimate an initial simple linear model of overall satisfaction. The objective is to examine the extent to which the different features associated with the shopping experience relate to the overall satisfaction reported by respondents.

Before modeling, the driver blocks used in the analysis were reorganized. This reorganization did not change the original variables or their values, but rather the way they were grouped for the modeling stage, so that the blocks would be more representative of the drivers they contain. For example, the original “Customer service” block combined variables related to staff interactions with variables associated with waiting times. It was therefore split into two blocks: “Staff and customer service” and “Waiting times and operational convenience.” In addition, some variables associated with special offers, such as the availability of products on offer and the appeal of the offers, were grouped into a separate block called “Offers and promotions.” Full details of the changes made are presented in Appendix C.1.

Likewise, modeling required the development of an imputation strategy for the problematic cells in the analysis dataset. These are cells containing values that disrupt the ordering (code 5: Don’t know/No answer) or no value at all (genuine and structural blanks). Since it is not possible to model using variables that disrupt the logic of the behavior of the phenomenon being studied, namely overall satisfaction, several alternatives for treating these cells were evaluated. The strategy used to address this problem is presented in depth in Appendix C.2.

Another change made before specifying the model was to reverse the original ordering of the responses to make the coefficients easier to interpret. As a result, both the response variable and the drivers follow an increasing order; each original value between 1 and 4 of feature i is replaced by $feature_{i,creciente} = 5 - feature_i$. The scale is thus oriented in ascending order: “Completely satisfied” is represented by the value 4, “Satisfied” by 3, “Slightly satisfied” by 2, and “Dissatisfied” by 1. From this point onward, this new ordering is maintained throughout the thesis.

Finally, the 25,117 observations from the 13 selected stores were filtered using the feature `ultima_visita` = less than 1 week. This filter was applied to reduce the possibility that respondents' recollections of their shopping experience were affected by recall bias [11]. Consequently, modeling was conducted using a dataset of 18,739 observations.

Once the modeling dataset had been consolidated, the decision was made to begin with the simplest possible linear model, including all features separately, the controls—gender and age by quantile—and an intercept.

$$SAT_i = \alpha + \sum_{k \in \mathcal{K}} \beta_k SP_{ki} + \gamma^\top DEM_i + \varepsilon_i$$

In the model:

- i indexes the individual respondent.
- SAT_i represents the continuous overall satisfaction in the interval $[1, 4]$ reported by individual i , where 4 is “Completely satisfied” and 1 is “Dissatisfied.”
- α represents the intercept.
- $\mathcal{K}$ represents the set of drivers included in the model.
- SP_{ki} represents the continuous evaluation in the interval $[1, 4]$ of driver $k \in \mathcal{K}$ given by individual i .
- β_k represents the marginal association between driver k and overall satisfaction, holding the other model variables constant. Since the drivers were centered, this coefficient is interpreted as the expected change in overall satisfaction for a one-unit increase in the driver relative to its mean.
- $\mathcal{DEM} = \{\text{age by quantile, gender}\}$ represents the set of demographic variables considered.
- DEM_i represents the vector of demographic variables for individual i .
- γ represents the vector of coefficients associated with the demographic variables.
- ε_i represents the model's error term.

In general terms, the coefficients in the model developed here make it possible to identify which dimensions of the in-store experience are most strongly associated with overall satisfaction, holding the other drivers and demographic controls constant. To make the coefficients more interpretable, the variables were centered, so that zero no longer represents a possible response on the scale, but rather the mean value of each feature. This is important for interpretation because the coefficients still measure the expected change in overall satisfaction for a one-unit increase in the driver, but this change is now interpreted relative to the variable's mean. This centering was essential because it also allowed structural blanks to be assigned a value of zero without treating them as an actual response. In those cases, the variable simply contributes no differential effect for that observation.

The coefficients were estimated using ordinary least squares. Since residual diagnostics showed evidence of heteroscedasticity, inference

is reported using HC3 robust standard errors. This correction does not change the estimated coefficients, but it does change the standard errors, confidence intervals, and associated p-values. The methodology is detailed in Appendix C.4.

To complement the interpretation of individual drivers, the analysis also examined whether effects were concentrated in particular blocks. To this end, a simple descriptive measure was constructed from the significant coefficients. Let $\mathcal{S}$ be the set of significant drivers in the model

$$\mathcal{S} = \{k : \hat{\beta}_k \text{ is significant} \}.$$

For each block b , the absolute weight was calculated as

$$W_b = \sum_{k \in \mathcal{S} \cap b} |\hat{\beta}_k|.$$

Then, to express this weight in relative terms, the following was calculated:

$$P_b = \frac{W_b}{\sum_{b' \in \mathcal{B}} W_{b'}}.$$

3.2. Imbalance in overall satisfaction and weighted models

The distribution of overall satisfaction has highly imbalanced categories. Most responses are concentrated in the high satisfaction levels, while the lower satisfaction categories are much less frequent, as shown in Figure 3.1.

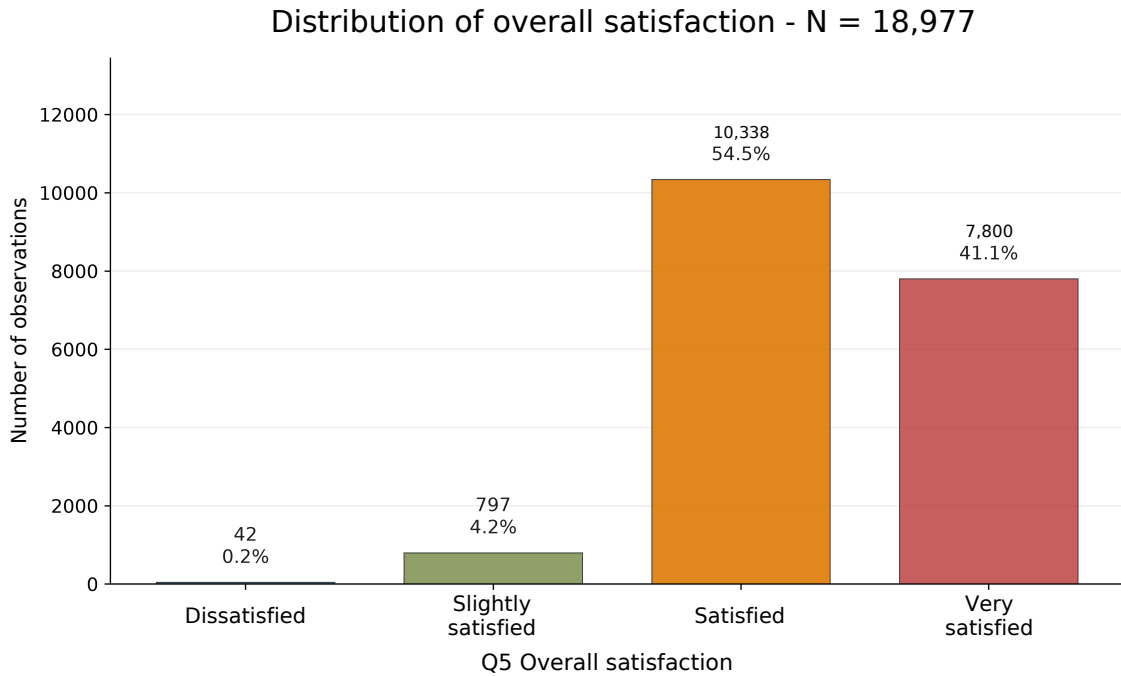


Fig. 3.1: Distribution of overall satisfaction in the training sample.

This imbalance poses a problem for estimating and interpreting the model, since an ordinary least squares fit tends to be dominated by the

most frequent categories. As a result, the model may perform well overall while providing a poorer explanation of the lower satisfaction cases, which are precisely those of greatest interest from an analytical perspective.

Consequently, weighted models were also estimated. The idea was to partially correct the weight of each category in the fit by assigning greater importance to observations belonging to less frequent categories. In this way, lower satisfaction cases are less diluted within a sample dominated by positive responses. The weights and the criterion used to assign them are reported in Appendix C.5.

Under this weighting criterion, two ways of grouping the overall satisfaction categories were considered. The first and most intuitive was to retain the original categories and use four different weights (WLS 4cat). However, as described above, this poses a problem because it gives too much weight to a category (1—“Dissatisfied”) that accounts for only 0.2% of observations. The result is a highly unstable model, with few relevant coefficients and very high standard deviations for most of them.

The decision was therefore made to group categories 1—“Dissatisfied” and 2—“Slightly satisfied” solely for calculating the model weights. This reduces the uncertainty that arose when category 1 was weighted separately, without removing those cases from the analysis. It is important to clarify that the target variable is not recoded onto a three-value scale: overall satisfaction continues to be modeled on its original scale from 1 to 4. The change is that observations with satisfaction levels 1 and 2 receive the same weight, corresponding to the low satisfaction group. Thus, weighting is defined over three ordered groups, $1 || 2 < 3 < 4$, while the regression continues to fit overall satisfaction as a numeric variable between 1 and 4. The drivers retain their original scale from 1 to 4.

A more stable model (WLS 3cat) with more interpretable coefficients was obtained (low p-values, narrower confidence intervals). This is shown in greater detail in the table in Appendix C.11.

3.3. Temporal cross-validation

On initial inspection, the weighted model appeared more reasonable. Nevertheless, it was appropriate to establish a criterion for comparing robustness across the three models. R^2 and adjusted R^2 are useful as initial measures of fit, but they are calculated on the same sample used to estimate the model. They are therefore insufficient to evaluate whether the model continues to behave reasonably when predicting observations that were not used during estimation.

The analysis also considered the quality of the model’s predictive ability as an indicator of the reliability of inference about the coefficients. Cross-validation (CV) was chosen for this purpose. Since overall satisfaction varies somewhat across survey waves, these were taken into account when splitting the data. For this reason, temporal CV was used instead of a random split. This decision is consistent with the work of Roberts et al. [16]: when data have a temporal structure, a random split may yield an overly optimistic evaluation. In this case, the ordering of survey waves was therefore preserved, training on earlier periods and validating on later periods.

The full results are presented in Appendix C.6. In summary, the initial unweighted model has the lowest overall errors, but the three-category model improves R^2 , mean absolute error (MAE), and mean squared error (MSE), adjusted and weighted by category, without exhibiting the instability of the four-category weighted model. For this reason, the three-category model was taken as the reference for interpretation.

3.4. Reducing drivers: PCA and feature selection

After selecting the three-category weighted model, the analysis evaluated whether the number of drivers could be reduced while maintaining performance similar to that of the full model. The latter includes 33 drivers, allowing a comprehensive interpretation of the experience, but it may also complicate inference because of the number of coefficients to analyze.

Two approaches to reducing model complexity were therefore considered. The first was to use PCA to summarize the drivers in a small number of principal components. The second was to use a feature selection method to retain a smaller subset of the original drivers. Both strategies reduce the number of variables, but they do not have the same interpretation. With PCA, the original drivers no longer appear directly in the model and are instead summarized in components. This can be useful from a statistical perspective because it allows the analysis to use very few variables, but it makes the results harder to interpret.

This matters for the objective of this thesis. The aim is to identify which specific aspects of the in-store experience are associated with overall satisfaction, using goodness of fit as a criterion for robustness. From the perspective of business interpretation, it is more useful to consider drivers individually or grouped into blocks that have a meaningful real-world interpretation (e.g., price, assortment, infrastructure, etc.) than principal components formed from combinations of drivers belonging to several different blocks.

Nevertheless, the analysis explored modeling with a number of principal components from a PCA of the data that explained approximately 50% of the variance. Since the original dataset of 13 physical stores had been modified through imputations and filters, a new PCA was performed on the sample. The first three principal components were used; together, they account for 50.33% of the explained variance.

The following model was estimated using these three components:

$$SAT_i = \alpha + \sum_{r=1}^R \delta_r PC_{ri} + \gamma^\top DEM_i + \varepsilon_i$$

where:

- PC_{ri} represents the value of principal component r for individual i .
- R represents the number of principal components included in the model. In this case, $R = 3$ is used.
- δ_r represents the association between principal component r and overall satisfaction, holding the other model variables constant.

The remaining terms are interpreted in the same way as in the baseline model. As already mentioned, unlike the coefficients associated with individual drivers, the coefficients δ_r do not represent the direct effect of a specific driver. Each principal component is a linear combination of several drivers. In other words, each component is constructed by combining the original drivers with different weights. These weights are called loadings and indicate how much each driver contributes to the construction of each principal component.

Since loadings show only how each component is constructed, rather than how that component relates to overall satisfaction, the aim was to translate the PCA model back into driver blocks. To do so, the estimated coefficients of each component were combined with the loadings of the drivers belonging to each block. This provides an approximation of each block's contribution within the model estimated using principal components. The details of the rationale for this combination are explained in Appendix C.7.1.

This model provides a less direct basis for interpreting specific drivers. It was therefore treated as an alternative that makes it possible to observe which broad dimensions of the experience are most strongly associated with overall satisfaction when the information is reduced to a few components, rather than as the definitive model for interpretation.

A second variable reduction strategy was therefore considered. Instead of replacing the drivers with principal components, this strategy seeks to select a subset of the original drivers. This reduces the number of variables in the model while retaining a direct interpretation of each coefficient.

SFS (Sequential Feature Selection) was used for this feature selection. This method was chosen because, among the alternatives evaluated, it produced the best results, with the lowest out-of-sample mean squared error calculated from the out-of-fold cross-validation predictions (MSE OOF) as the main criterion. It also retained an intermediate number of drivers: fewer than the full model, but not so few as to lose useful information about the in-store experience. Further details are provided in Appendix C.7. A grid search was performed over the SFS tolerance constraint parameter (`tol`), reaching an optimal number of coefficients $K = 20$ under the constraint `tol` = 0.0001, with an error of RMSE OOF ≈ 0.540 (the square root of MSE).

Comparing the model reduced through SFS with the full model of 33 drivers shows that reducing the number of variables does not substantially change the overall interpretation. This is summarized in Table 3.1.

Tab. 3.1: Comparison between the full weighted model and the model reduced through SFS.

Criterion	Result
Significant drivers in the full WLS model	16
Significant drivers in the full WLS model retained by SFS	16 of 16
Significant drivers among the top 15 that remain significant in SFS	13 of 13
Top 10 by absolute magnitude in the full WLS model retained by SFS	10 of 10
Top 15 by absolute magnitude in the full WLS model retained by SFS	14 of 15
Coefficient sign preserved in SFS	19 of 20
Correlation of magnitude rankings between the two models	0.983

For the reasons discussed above, the WLS 3cat model with SFS selection and $K = 20$ drivers was adopted as the main model for inference. This choice allows the analysis to use a smaller model than the one with 33 drivers without losing the substantive interpretation of the coefficients.

3.4.1. Interpreting the coefficients

Since the drivers were centered, the coefficients should be interpreted as associations between deviations from a driver's mean and overall satisfaction. That is, a positive coefficient indicates that an evaluation of the driver above its mean is associated with higher overall satisfaction, holding the other variables constant. In turn, structural blanks were set to zero, so they contribute neither a positive nor a negative deviation from the driver's mean.

3.5. A model of satisfaction as an ordinal variable

In the previous model, the drivers were treated as centered numeric variables. This modeling approach assumes that changes between categories have a relatively homogeneous effect on satisfaction. This assumption may be problematic because the responses come from an ordinal scale. In other words, moving from a low evaluation to an intermediate evaluation does not necessarily have the same effect as moving from an intermediate evaluation to a high evaluation. For this reason, a second model was estimated in which the drivers no longer enter as a single numeric variable and are instead represented by indicator variables.

In this model, category 3—"Satisfied" is taken as the reference. For each driver, a distinction is then made between low evaluations (1||2) and high evaluations (4). This allows the model to compare separately how each type of response is associated with overall satisfaction, without imposing a single linear relationship over the entire range of the scale.

Consequently, the model is defined as follows:

$$SAT_i = \alpha + \sum_{k \in \mathcal{K}} \left(\beta_k^- D_{ki}^{1||2} + \beta_k^+ D_{ki}^4 \right) + \gamma^\top DEM_i + \varepsilon_i$$

where:

- i indexes the individual respondent.
- SAT_i represents the continuous overall satisfaction in the interval $[1, 4]$ reported by individual i , where 4 is "Completely satisfied" and 1 is "Dissatisfied."
- α represents the intercept.
- $\mathcal{K}$ represents the set of drivers included in the model.
- For each driver k , category 3 is taken as the reference category. In other words, the estimated coefficients are always interpreted relative to an intermediate evaluation of the driver.

- $D_{ki}^{1||2}$ is an indicator variable that takes the value 1 if individual i evaluated driver k in categories 1 or 2, and 0 otherwise. This variable represents a low evaluation of the driver.
- D_{ki}^4 is an indicator variable that takes the value 1 if individual i evaluated driver k in category 4, and 0 otherwise. This variable represents a high evaluation of the driver.
- β_k^- represents the expected difference in overall satisfaction associated with evaluating driver k as 1||2, compared with evaluating it as 3, holding the other model variables constant.
- β_k^+ represents the expected difference in overall satisfaction associated with evaluating driver k as 4, compared with evaluating it as 3, holding the other model variables constant.
- $\mathcal{DEM} = \{\text{age by quantile, gender}\}$ represents the set of demographic variables considered.
- DEM_i represents the vector of demographic variables for individual i .
- γ represents the vector of coefficients associated with the demographic variables.
- ε_i represents the model's error term.

The same imputation and data filtering criteria were maintained.

In this model, each driver appears twice: once through the coefficient associated with a low evaluation (1||2), and once through the coefficient associated with a high evaluation (4—“Completely satisfied”). Since the model includes an intercept, there are 66 dummy variables. In both cases, the comparison is made against category 3—“Satisfied,” which serves as the reference ($\beta_k^{\text{Satisfecho}} = 0$ for every $k \in \mathcal{K}$).

Starting from the intuition that a better evaluation of a feature should be associated with higher overall satisfaction and a worse evaluation with lower satisfaction, the analysis first examined whether the estimated coefficients had the expected sign relative to the reference category. However, this initial reading should be treated as purely descriptive, since in some cases the confidence intervals include zero.

For this reason, a more restrictive criterion was adopted to identify drivers with a more stable relationship relative to the reference category. Cases were flagged when the coefficient associated with a high evaluation was positive, the coefficient associated with a low evaluation was negative, and neither 95% confidence interval included zero. In these cases, the model provides stronger support for the claim that a high evaluation is positively associated with overall satisfaction, while a low evaluation is negatively associated with it, always relative to the reference value $\beta_k^{\text{Satisfecho}}$.

3.5.1. A loss aversion measure for asymmetry between low and high evaluations

The idea of loss aversion refers to a possible asymmetry between losses and gains. In the context of customer satisfaction, this means that a poor evaluation of a driver may carry more weight than a good evaluation of the same driver. In other words, the aim is not only

to determine whether a low evaluation reduces satisfaction and a high evaluation increases it, but also to compare whether the decrease associated with a low evaluation is greater than the increase associated with a high evaluation.

Building on the previous calculations, a comparison measure was added to assess whether, for a given driver, the decrease associated with a low evaluation is greater than the increase associated with a high evaluation.

For each driver k , the following measure was defined:

$$\Delta_k = |\beta_{\text{baja},k}| - \beta_{\text{alta},k}.$$

Since, in the expected cases, $\beta_{\text{baja},k} < 0$ and $\beta_{\text{alta},k} > 0$, this expression can also be written as

$$\Delta_k = -\beta_{\text{baja},k} - \beta_{\text{alta},k}.$$

If $\Delta_k > 0$, the penalty associated with a low evaluation is greater than the reward associated with a high evaluation. Conversely, if $\Delta_k < 0$, the opposite holds: the positive effect has a greater magnitude than the negative effect. Thus, this measure provides a simple summary of the comparison between the two coefficients. However, its value must be interpreted with care. A positive Δ_k does not automatically imply statistical evidence of loss aversion. Reaching that conclusion also requires evaluating whether the difference between the two magnitudes is statistically different from zero. This is important because the model may show that a high evaluation adds to satisfaction and a low evaluation subtracts from it, while the difference between the two effects is not large enough to be considered significant.

To conclude that the asymmetry is statistically significant, it is necessary to test directly whether Δ_k differs from zero; this is explained in greater detail in Appendix C.8.

3.5.2. Robustness check using SFS

As a robustness criterion, a reduced version of the model was also estimated using SFS. In this case, selection was performed at the driver level rather than on individual dummy variables. That is, if a driver was selected, both of its indicator variables entered the model: β_k^+ and β_k^- . This is important because comparing the positive and negative effects only makes sense if both coefficients are present for the same driver.

The aim was to observe whether drivers with a signal consistent with loss aversion were retained after reducing the number of variables. To do so, a grid search was performed over different values of the SFS tolerance parameter (`tol`). The criterion was to minimize out-of-sample error. As a result, $K = 15$ drivers were selected for `tol` = 0.0001, with an RMSE OOF ≈ 0.528 .

It should be clarified that SFS does not select drivers according to their degree of asymmetry, but according to their contribution to the out-of-sample mean squared error (MSE OOF) of CV for the model under consideration. Therefore, a driver may show a clear signal of loss aversion in the full model but not be selected by SFS if its improvement in terms of error is very small or negative.

For this reason, the reduced model is used as a robustness check, while the main interpretation of loss aversion is based on the full model. For a more extensive analysis, see Appendix C.8.

5. CONCLUSIONS

This thesis develops an analytical framework for identifying, quantifying, and evaluating the determinants (drivers) of customer satisfaction in a supermarket chain, covering the period from June 2021 to March 2026. Throughout the study, a critical and progressive analytical strategy was developed, with a sequence of steps aimed at evaluating different aspects of the problem.

The starting point of this research addresses the fragmentation of the original information, initially comprising more than 34,000 surveys collected using four structurally different questionnaire versions. In response to this situation, questions and variables were unified through equivalence relationships and the implementation of the Multiple Imputation by Chained Equations (MICE) algorithm. This technique made it possible to treat non-evaluative responses and missing data consistently. Together with the regional restriction to 13 stores with complete temporal coverage and the application of last-visit control filters to reduce recall bias, this allowed a final modeling dataset of 18,739 observations to be consolidated.

The subsequent characterization and exploration of the dataset revealed two major challenges: a markedly asymmetric distribution of overall satisfaction, strongly imbalanced toward outstanding ratings; and high distributional redundancy and internal correlation among the operational attributes of infrastructure, service, and quality, also observed through principal component analysis. This diagnosis showed that direct estimation using standard linear regression tools based on traditional least squares could lead to interpretation problems due to collinearity and the imbalance in the target variable. It was therefore necessary to reconfigure the inferential strategy toward models capable of accounting for this complexity.

In response to this need, the modeling strategy progressed from the simplest linear formulation using Ordinary Least Squares (OLS), which served as a baseline for comparison. Once its limitations became apparent, a transition was made to a Weighted Least Squares (WLS) scheme designed to compensate for the imbalance in the target variable, while integrating a Sequential Feature Selection (SFS) algorithm in parallel. The validity of each methodological change was tested empirically through a temporal cross-validation scheme. These results showed that linear optimization made it possible to isolate a parsimonious core of $K = 20$ interpretable variables without degrading the model's ability to generalize.

One contribution of the thesis was to avoid assuming that satisfaction drivers necessarily act in a linear and symmetric way. By allowing a comparison between low and high evaluations of the same attribute, evidence consistent with loss aversion was found for some drivers. These results suggest that, in the sample analyzed, a poor evaluation of certain attributes may penalize overall satisfaction more than a good evaluation increases it. Nevertheless, this interpretation does not imply that all drivers always follow this logic; rather, relevant asymmetries were identified in some cases.

Finally, segmenting the modeling by store format showed that

consumer sensitivity does not appear to be fully shared or homogeneous across the chain.

As a general conclusion, this thesis shows that careful treatment of the dataset and a progressive modeling strategy make it possible to develop a richer interpretation of customer satisfaction. The results not only identify average satisfaction drivers, but also reveal possible asymmetries in some attributes and relevant differences by store format.

Appendix A

BUILDING THE UNIFIED DATASET

A.1. Questionnaire types and administration periods

This section describes the process of merging the survey types to obtain a unified dataset. It first presents how the surveys were classified by questionnaire type. This classification was not defined arbitrarily, but was based on the information provided by the company about changes to the form over time. In other words, a survey belongs to one type or another according to the survey wave in which it was conducted, as shown in Table A.1.

Table A.1: Questionnaire types by administration period.

Questionnaire type	Administration period
Type 1	June 2021–August 2021
Type 2	September 2021–April 2024
Type 3	May 2024–February 2026
Type 4	March 2026 onward

This distinction matters because each questionnaire type has its own structure. Some questions remain unchanged over time, others change position, some appear only in certain types, and there may also be different response routes within the same questionnaire. For this reason, the files could not be concatenated directly without first undergoing a unification process.

Merging the 40 available files was necessary to make proper use of the samples. The differences between questionnaire types, and even between questionnaires within the same type (changes between survey waves), revealed the need to unify them.

This unification process is described and formalized step by step below.

A.2. Collapsing equivalent questions

First, this section describes how the existence of different routes within the questionnaires affected the structure of the original variables.

Differences between branches in terms of size and available services explain why the questionnaire did not have a single structure for all respondents. Consequently, not everyone answered the same set of questions within the same questionnaire. The route through the form depended in part on the branch where each respondent had made their most recent purchase. In this context, the same aspect being evaluated could appear in different versions of a question: these shared several subquestions, but differed in whether they included or excluded certain aspects associated with services available only at some branches.

This procedure was applied to all samples, regardless of questionnaire type.

Figure A.1 provides a schematic summary of this procedure. Its formalization is presented below.

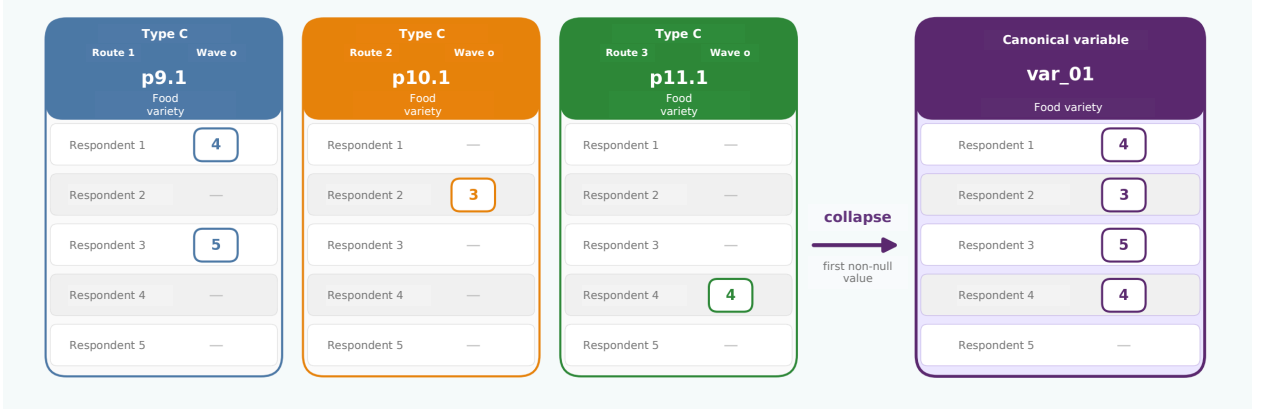


Figure A.1: Example of collapsing equivalent questions located on different questionnaire routes into a single canonical variable.

A.2.1. Formalization

Let c be a questionnaire type, and let I_c denote the total number of questions it contains. The question index i therefore takes values in $\{1, \dots, I_c\}$.

Given a question i in c , define the set

$$\mathcal{P}_{c,i} = \{P_{c,i,j} : j = 1, \dots, J_{c,i}\},$$

where $P_{c,i,j}$ represents subquestion j associated with question i of questionnaire type c , and $J_{c,i}$ represents the total number of subquestions associated with that question.

For simple questions, that is, questions that are not subdivided, set $J_{c,i} = 1$. These questions are represented by a single element $P_{c,i,1}$.

Also define an equivalence relation $\sim$ between original subquestions. We say that

$$P_{c,i,j} \sim P_{c,\ell,m}$$

if both subquestions refer to the same aspect being evaluated, even when they belong to different questions or routes within the same questionnaire type c .

Thus, for each final aspect d to be constructed in the unified dataset—for example, food variety, branch cleanliness, or meat quality—a set of equivalent subquestions can be defined as

$$\mathcal{E}_{c,d} = \{P_{c,i_1,j_1}, P_{c,i_2,j_2}, \dots, P_{c,i_k,j_k}\},$$

such that all its elements are equivalent to one another under the relation $\sim$.

Let $U_{o,c}$ be the set of observations corresponding to a survey wave o associated with questionnaire type c . Each observation $u \in U_{o,c}$ has an associated branch s_u , where the respondent made their most recent purchase at the chain, and follows a unique route r_{c,s_u} within questionnaire type c .

For each observation $u \in U_{o,c}$, let $Y_{c,i,j}^{(u)}$ denote the response recorded for subquestion $P_{c,i,j}$.

In this context, r_{c,s_u} is defined as the set of subquestions belonging to the route corresponding to observation u . Therefore, a response $Y_{c,i,j}^{(u)}$ can only be defined if $P_{c,i,j} \in r_{c,s_u}$.

Given a set of observations $\mathcal{O}_{c,o,d} \subseteq U_{o,c}$ in wave o , associated with the final aspect d , with

$$\mathcal{O}_{c,o,d} = \{u_{h_1}^{(o)}, u_{h_2}^{(o)}, \dots, u_{h_n}^{(o)}\},$$

for every $u_{h_t}^{(o)} \in \mathcal{O}_{c,o,d}$, there exists $P_{c,i_t,j_t} \in \mathcal{E}_{c,d} \cap r_{c,s_{u_{h_t}^{(o)}}}$ such that $u_{h_t}^{(o)}$ answers

P_{c,i_t,j_t} , with $Y_{c,i_t,j_t}^{(u_{h_t}^{(o)})}$ being the corresponding response.

Because these are distinct observations, they occupy different rows in the dataset. In turn, the subquestions contained in $\mathcal{E}_{c,d}$ correspond to different columns. Since routes within the same questionnaire type are mutually exclusive, no more than one subquestion in $\mathcal{E}_{c,d}$ can belong to the route r_{c,s_u} of a given observation $u \in U_{o,c}$. That is, if $P_{c,i_a,j_a}, P_{c,i_b,j_b} \in \mathcal{E}_{c,d}$ and also $P_{c,i_a,j_a}, P_{c,i_b,j_b} \in r_{c,s_u}$, then necessarily $a = b$.

Under this condition, for each observation $u_{h_t}^{(o)} \in \mathcal{O}_{c,o,d}$, define the collapsed value as

$$V_{o,d}^{(u_{h_t}^{(o)})} = Y_{c,i_t,j_t}^{(u_{h_t}^{(o)})}.$$

If, for an observation $u \in U_{o,c}$, there is no defined response $Y_{c,i,j}^{(u)}$ associated with a subquestion $P_{c,i,j} \in \mathcal{E}_{c,d}$, the corresponding value remains null. Thus, this procedure neither modifies nor imputes responses; it only reorganizes equivalent responses from different routes into a single variable.

The collapsed column associated with aspect d in wave o is given by

$$V_{o,d} = \left(V_{o,d}^{(u)} \right)_{u \in U_{o,c}}.$$

Finally, consider the set of final aspects defined for survey wave o , $\mathcal{D}_o = \{d_1, d_2, \dots, d_{p_o}\}$. For each $d_r \in \mathcal{D}_o$, the variable V_{o,d_r} represents the column of the collapsed dataset associated with aspect d_r .

$$\mathcal{B}_o^{\text{col}} = [V_{o,d_1} \mid V_{o,d_2} \mid \dots \mid V_{o,d_{p_o}}].$$

Thus, $\mathcal{B}_o^{\text{col}}$ represents the collapsed dataset corresponding to wave o , constructed by concatenating all the columns $V_{o,d}$ associated with the final aspects evaluated in that wave.

A.3. Building a common structure across questionnaire types

Once equivalent questions had been collapsed within each survey wave, the next step was to bring the variables from the different questionnaire types into a

common structure. For this stage, type 4 was chosen as the reference because it was the most recent version. This version resulted from various modifications that the company had considered appropriate over time. Consequently, it was adopted as the most representative reference structure among the four types currently available.

In terms of structural complexity and number of questions, the questionnaires were found to have become simpler over time. Type 4 is therefore a shorter version of the earlier types, particularly compared with types 1 and 2. Type 3 is practically identical to type 4, except for some details related to promotions, which is to be expected given the dynamic nature of this type of content.

For this reason, the questions and subquestions in type 4 were taken as the starting point, and their presence in the other questionnaire types was assessed. This defined a final core of variables that were comparable over time. Excluding the blocks associated with Marketing/Advertising, a main core was identified relating to overall satisfaction, price, quality, variety, value for money, competition, and demographics. This core persisted throughout the period analyzed, although at different levels of detail depending on questionnaire type.

Given the similarity between types 3 and 4, this process is simpler in that case. Types 1 and 2, however, required more careful treatment because of their greater length and structural differences relative to the more recent questionnaires.

Figure A.2 provides a schematic summary of this procedure. Its formalization is presented below.

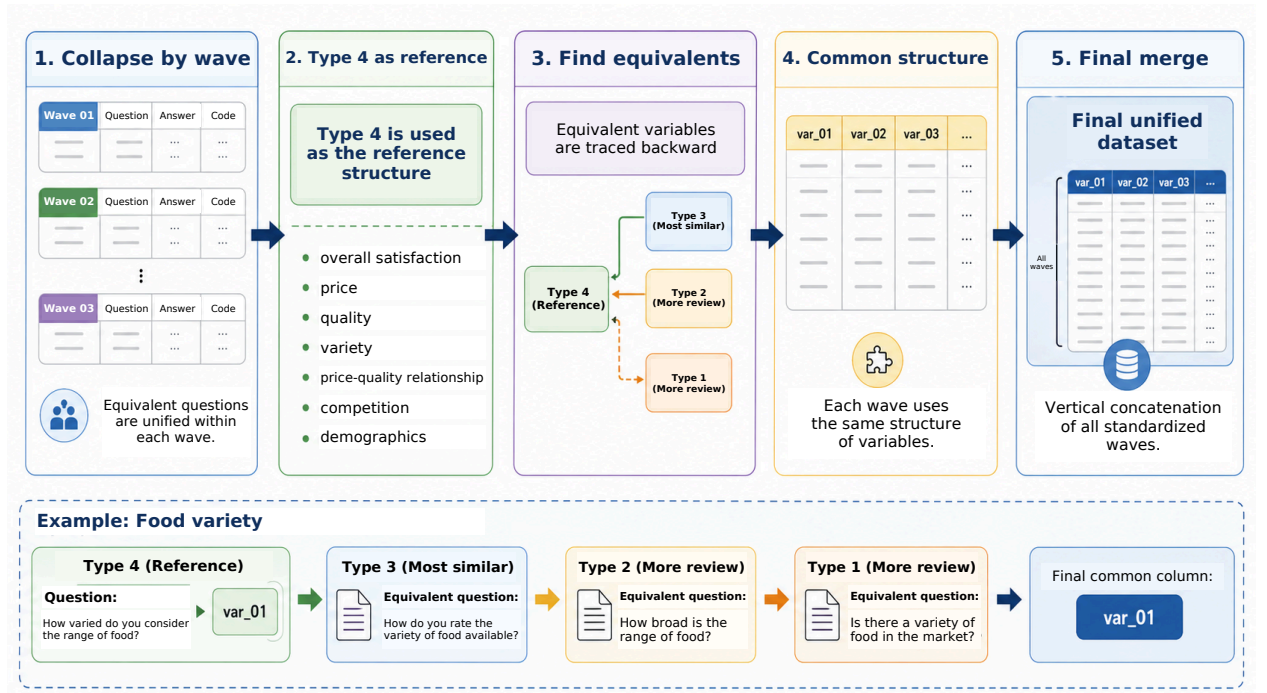


Figure A.2: Diagram of the construction of a common structure across questionnaire types using type 4 as the reference.

A.3.1. Formalization

Let o be a wave and $\mathcal{B}_o^{\text{col}}$ its collapsed dataset.

Define

$$\mathcal{D}_4 = \{d_1, d_2, \dots, d_q\}$$

as the set of aspects evaluated in the type 4 questionnaire, and let $\mathcal{D}_4^{\text{ref}}$ be those aspects considered relevant to the analysis, such that $\mathcal{D}_4^{\text{ref}} \subseteq \mathcal{D}_4$.

Consider $\mathcal{D}_c$ for each questionnaire type $c \neq 4$.

Each $d \in \mathcal{D}_c$ already has an associated set $\mathcal{E}_{c,d}$, defined above, and can therefore generate a collapsed column $V_{o,d}$ in each wave o for that type.

For $c \neq 4$, define the correspondence function

$$\phi_c : \mathcal{D}_4^{\text{ref}} \rightarrow \mathcal{D}_c \cup \{\emptyset\}.$$

This function maps an aspect $d_a \in \mathcal{D}_4^{\text{ref}}$ to $d_c \in \mathcal{D}_c$ if a correspondence exists ($\phi_c(d_a) = d_c$), or to the empty set if it does not ($\phi_c(d_a) = \emptyset$).

In the dataset, if wave o belongs to questionnaire type c , a new standardized column W_{o,d_a} is defined for each aspect d_a such that $W_{o,d_a} = V_{o,d_c}$. In other words, the corresponding collapsed column in that wave is taken and brought into the reference structure.

If $\phi_c(d_a) = \emptyset$, then W_{o,d_a} remains a null column for that wave.

Finally, all waves have the same structure, and the aspects evaluated, d_a , become features or variables:

$$\mathcal{B}_o^{\text{std}} = [W_{o,d_1} \mid W_{o,d_2} \mid \dots \mid W_{o,d_k}]$$

where

$$\mathcal{D}_4^{\text{ref}} = \{d_1, d_2, \dots, d_k\}$$

with $k \leq q$.

The final dataset is then obtained by vertically concatenating the standardized waves:

$$\mathcal{B}^{\text{final}} = \text{concat} \left(\mathcal{B}_{o_1}^{\text{std}}, \mathcal{B}_{o_2}^{\text{std}}, \dots, \mathcal{B}_{o_T}^{\text{std}} \right).$$

Appendix C

METHODOLOGICAL DECISIONS FOR MODELING

C.1. Reorganization of driver blocks

This section details the reorganization of blocks carried out before the modeling stage. As mentioned in the main section, this reorganization did not change the values of the original variables, but rather the way the drivers were grouped to make the models easier to interpret.

In some cases, the change consisted only of renaming blocks to make them clearer. For example, the “Quality” block became “Product quality,” and the “Infrastructure” block became “Environment and infrastructure.” In other cases, classification changes were made because some original blocks combined variables that could be interpreted better if separated into distinct dimensions.

The most important change was the separation of the original “Customer service” block. This block combined variables related to staff interactions with variables associated with waiting times. To improve interpretation, it was split into two dimensions: “Staff and customer service” and “Waiting times and operational convenience.” This may make sense when considering that staff friendliness or availability does not necessarily measure the same thing as waiting at the checkout, butcher counter, or deli counter.

The “Offers and promotions” block was also defined, grouping variables related to the availability and attractiveness of offers. These variables had originally been distributed across blocks such as “Assortment / availability” and “Prices,” but for modeling it was clearer to treat them as a dimension in their own right.

Table C.1: Comparison between original and final blocks.

Original block	Final block	Type of change
Assortment / availability	Assortment and availability	Renaming
Quality	Product quality	Renaming
Prices	Price	Renaming
Value for money	Perceived value / value for money	Conceptual renaming
Infrastructure	Environment and infrastructure	Renaming
Customer service	Staff and customer service	Substantive reassignment
Customer service	Waiting times and operational convenience	Substantive reassignment
Assortment / availability	Offers and promotions	Substantive reassignment
Prices	Offers and promotions	Substantive reassignment

Most changes consisted of renaming blocks to make them clearer. However, there were also some classification changes. These involved variables that it was more reasonable to place in another block for the modeling stage, because the new grouping better represents what those variables measure.

In particular, the change from “Infrastructure” to “Environment and infrastructure” sought to make

the block name more representative of the variables it contains. Some variables, such as shopping carts and baskets or restrooms, clearly refer to physical elements of the store, but they also form part of the customer’s overall experience at the branch. Thus, the name “Environment and infrastructure” better captures this dual dimension: the physical aspects of the space, on the one hand, and how those elements affect the comfort of the shopping experience, on the other.

Table C.2: Variables reassigned between blocks.

Variable	Feature	Original block	Final block
var_05	Availability of products on offer	Assortment / availability	Offers and promotions
precio_07	Attractiveness of offers	Prices	Offers and promotions
atencion_01	Staff availability	Customer service	Staff and customer service
atencion_06	Staff friendliness	Customer service	Staff and customer service
atencion_02	Waiting time at the butcher counter	Customer service	Waiting times and operational convenience
atencion_03	Waiting time at the deli counter	Customer service	Waiting times and operational convenience
atencion_04	Waiting time at the prepared food counter	Customer service	Waiting times and operational convenience
atencion_05	Waiting time at the checkout	Customer service	Waiting times and operational convenience

The most relevant separation was that of the “Customer service” block. In the original classification, this block combined variables associated with staff interactions and variables associated with waiting times. In the final classification, these dimensions were separated into “Staff and customer service” and “Waiting times and operational convenience.” This may make sense when considering that staff friendliness or availability does not necessarily measure the same thing as waiting at the checkout, butcher counter, or deli counter.

In addition, the “Offers and promotions” block was created to bring together variables specifically associated with the availability and attractiveness of offers. This distinguishes them from the general assortment and price level, which represent related but not identical dimensions.

C.2. Treatment of missing values and code 5

Once the unified dataset had been obtained, the features to be used as explanatory variables had been selected, and overall satisfaction had been defined as the response variable, an important problem arose for modeling. In some cases, respondents answered “Don’t know/No response” or “Not applicable,” while in others the cell was simply empty. Before proceeding to the modeling stage, it was therefore necessary to distinguish these situations and define how to handle them.

Each of these cases represents a different problem and requires a separate analysis to decide how to handle it.

It is important to clarify that, for the type of model used, drivers are treated as continuous variables. Consequently, the values imputed using MICE were not rounded.

They were kept within the valid range of the scale, but as continuous values. Rounding them could lose some of the information generated by MICE and unnecessarily force integer values.

C.2.1. Characterization of problematic cells

First, a problem associated with the loss of important information during the merge had to be resolved. Blanks, or empty cells, were categorized into two types: genuine blanks and structural blanks. The former correspond to cases in which respondents did not answer even though they were supposed to, given their route through the questionnaire. The latter correspond to cases in which respondents were not supposed to answer because the subquestion belonged to a route that did not apply to them.

During the merge, this distinction is lost at the stage of collapsing variables within a wave. If a respondent did not answer because the question did not apply to them (e.g., the branch they visited does not have a restroom), the final variable will be blank after collapsing.

The same occurs if a respondent does not answer when they are supposed to: the variables are collapsed and the cell remains blank.

This poses a problem because, once the variables have been collapsed, there is no way to distinguish a relevant empty cell, that is, a genuine blank, from one that was not applicable because it belonged to a different route. To resolve this problem, a dictionary was used, based on knowledge of the route that each branch must follow within each questionnaire type. This makes it possible to distinguish problematic empty cells from irrelevant ones. Here, “irrelevant” refers to their lack of contribution to characterizing the observation. Even so, they also require special treatment because they form part of the dataset.

Code 5 posed a problem of a different nature: it breaks the ordinal scale, since it is not comparable in order with any other valid response.

Table C.1 shows how many blank cells, structural blanks, and code 5 cells there are in the final dataset.

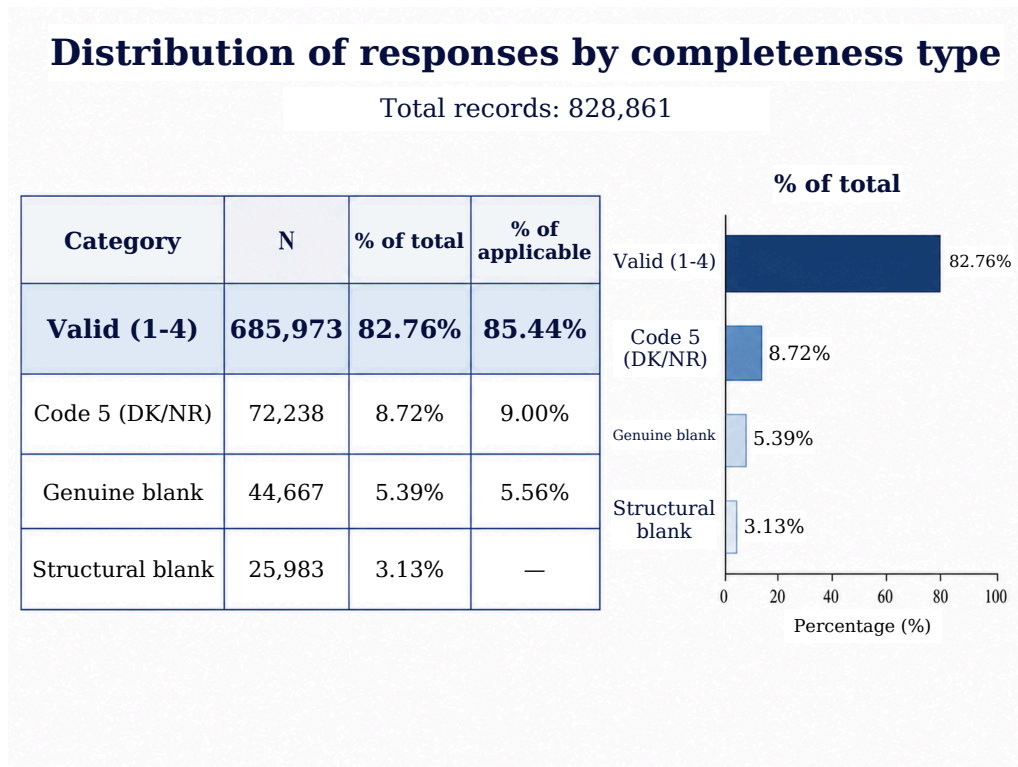


Figure C.1: Percentage of problematic cells.

Given this context, a more detailed analysis of the cases of interest—code 5 and genuine blanks—was carried out. Figures C.2 and C.3 show in greater detail how many problematic cells there are by survey type and for each observation.

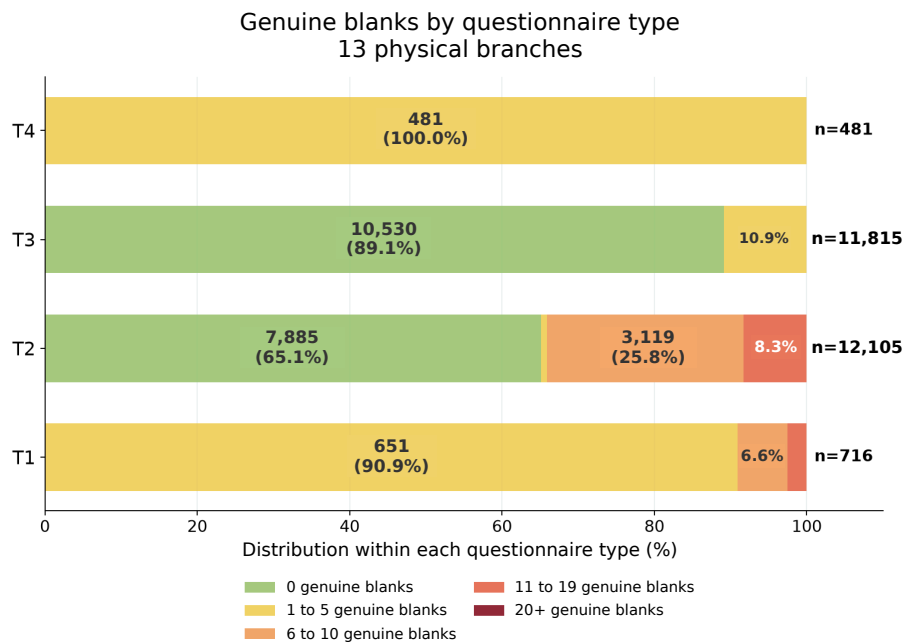


Figure C.2: Number of blanks by survey type and sample.

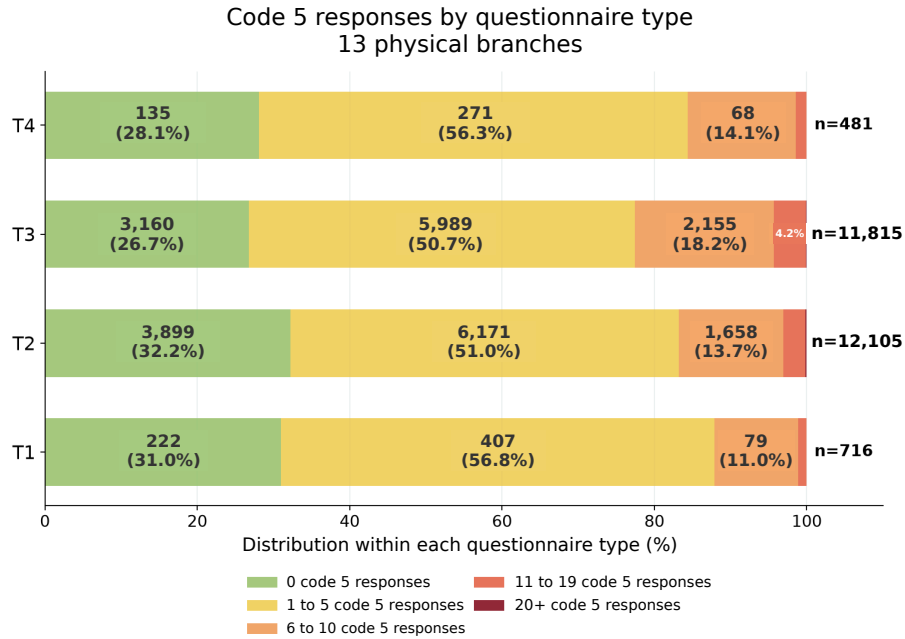


Figure C.3: Number of code 5 responses by survey type and sample.

These charts showed that a very large percentage of observations contain problematic cells. Consequently, adopting some form of imputation strategy appeared reasonable. Since observations containing these types of cells cannot be used directly in the model, excluding them from the training sample would substantially reduce the sample size. It could also bias the sample, since the presence of these cells might be related to the other variables in the observation or to variables beyond the scope of the data available to us.

The next step was to examine whether these cells displayed a random pattern across variables.

An uneven distribution was observed for both genuine blanks and code 5. The presence of these cells in the observations does not appear to be completely random. These patterns are therefore incompatible with an MCAR (Missing Completely At Random) mechanism.

Variables associated with waiting times are overrepresented among genuine blanks. For code 5, by contrast, there is overrepresentation among more specific variables, that is, variables that are not universal across branches or that depend on the use of certain departments. This suggests that the two types of problematic cells do not necessarily arise from the same mechanism.

The same was observed when analyzing their distribution by branch.

This analysis was important because it shows that problematic cells are not distributed completely at random. However, the observed data do not allow a conclusive determination of whether the mechanism is MAR (Missing At Random) or MNAR. In this case, the MAR assumption should be interpreted as a working hypothesis needed to apply imputation methods, not as a property that

can be verified directly from the data.

This distinction is relevant because code 5 responses and genuine blanks appear to follow different patterns. Code 5 responses are more concentrated in variables associated with specific or optional sections of the store, which may be linked to respondents' actual use of those sections. Genuine blanks, by contrast, show a more heterogeneous pattern that could combine omission, non-use, response fatigue, avoidance, or recording errors. For this reason, imputation was not treated as a single definitive solution, but as a methodological decision that could be compared with alternative strategies in the future.

C.3. Imputation strategy for problematic cells

This section formally presents the methods and criteria used to impute problematic cells. It also details the criterion used to decide which imputation strategy is most appropriate.

Based on the preceding analysis, MICE was used as an imputation strategy under a working MAR assumption. The idea is that, at least in part, the problematic values in an observation may be related to the other available features. MICE makes use of this information by imputing each variable conditional on the other observed variables. This procedure is explained in detail in Section C.3.1.

However, since the observed patterns do not guarantee that the MAR assumption holds fully, simpler alternative strategies were also compared, such as imputation using the median or mean, either of the block or of the variable. These strategies were also compared against the pairwise correlation matrix and the complete-case scenario. Thus, the final imputation strategy is interpreted as the result of a sensitivity analysis, rather than a strong assertion about the true mechanism generating the missing values.

The criterion used to choose the imputation strategy was based on comparing correlation matrices. The pairwise Spearman correlation matrix, R_S^P , was taken as the reference and compared with the Spearman correlation matrix obtained after applying each imputation strategy E , R_S^E .

Spearman was chosen instead of Pearson because most of the features come from ordinal categorical scales. That is, although responses are represented numerically, what matters most is the order of the categories and not necessarily the exact distance between them. For this reason, Spearman is more appropriate, since it calculates association from the ranks of the variables.

C.3.1. MICE (Multiple Imputation by Chained Equations)

MICE was used to impute missing values, following the general logic of multiple imputation through a sequence of regression models proposed by Raghunathan et al. (2001) [17].

Let

$$\mathcal{D}_4^{\text{ref}} = \{d_1, d_2, \dots, d_k\}$$

be the set of features.

Let $u \in \mathcal{B}^{\text{final}}$ be a sample, and let X_{u,d_j} be the value associated with feature d_j to be imputed for u .

If

$$d_{-j} = \{d_1, \dots, d_{j-1}, d_{j+1}, \dots, d_k\} = \mathcal{D}_4^{\text{ref}} \setminus \{d_j\},$$

then d_{-j} represents the set of all features except d_j . We define

$$X_{u,d_{-j}} = \{X_{u,d_h} : d_h \in d_{-j}, X_{u,d_h} \sim F_{d_h}\}$$

as the set of features of u with feasible values. That is, $X_{u,d_h} \sim F_{d_h}$, where F_{d_h} is the distribution associated with feature d_h .

The idea of MICE is to estimate, for each feature with values to be imputed, a conditional distribution of the form

$$\widehat{\mathbb{P}}(X_{u,d_j} \mid X_{u,d_{-j}}).$$

This distribution approximates the values that X_{u,d_j} could take, given the other feasible values available for that observation.

The procedure is applied sequentially. That is, one feature is imputed using the other available features, those imputed values are then updated, and the procedure continues with another feature. This process is repeated iteratively, forming a chain of conditional equations.

In multiple imputation, this procedure is repeated for $\ell = 1, \dots, m$. Each repetition yields a different imputed dataset. Therefore, for a value X_{u,d_j} to be imputed, the procedure does not yield a single final value, but one possible realization in each imputed dataset:

$$\widehat{X}_{u,d_j}^{(\ell)} \sim \widehat{\mathbb{P}}(X_{u,d_j} \mid X_{u,d_{-j}}), \quad \ell = 1, \dots, m.$$

The parameter m represents the number of imputed datasets generated. Thus, the procedure does not produce a single completed version of the dataset, but a set of m imputed datasets,

$$\mathcal{B}^{(1)}, \mathcal{B}^{(2)}, \dots, \mathcal{B}^{(m)}.$$

Each of these datasets contains one possible version of the missing values. In this way, multiple imputation incorporates part of the uncertainty associated with values that were not observed.

The model is then estimated separately on each imputed dataset. If $\widehat{\theta}^{(\ell)}$ represents the estimator obtained from imputed dataset ℓ , the combined point estimate can be written as

$$\bar{\theta} = \frac{1}{m} \sum_{\ell=1}^m \widehat{\theta}^{(\ell)}.$$

However, the final inference is not based solely on this average. Variability within each imputation and variability between imputations are also considered, following the logic of Rubin's rules. In other words, multiple imputation not only averages results, but also incorporates the uncertainty generated by completing unobserved values.

In this study, the selected strategy was applied in two stages. First, applicable genuine blanks were imputed using MICE with $m_{\text{blanks}} = 2$. Then, codes 5/99 were imputed using MICE with $m_5 = 2$. Therefore, the total number of imputed datasets used to combine the results was

$$m_{\text{total}} = m_{\text{blanks}} \times m_5 = 2 \times 2 = 4.$$

Consequently, unlike single imputation, this procedure does not depend on just one possible reconstruction of the dataset, but considers different feasible versions of the missing values.

C.3.2. Criterion for selecting among imputation strategies

The pairwise correlation matrix was chosen as the reference because it preserves the correlation between the raw variables with a greater degree of accuracy. For this matrix, blanks and code 5 responses are converted to NaN.

Let

$$\mathcal{S} = \{(d_j, d_k) : d_j, d_k \in \mathcal{D}_4^{\text{ref}}, 1 \leq j < k \leq K\}.$$

be the set of final features.

Consider two correlation matrices: R^P , the pairwise correlation matrix of a sample $\mathcal{M}$, and R^E , the correlation matrix of the same sample imputed using strategy E .

For a pair of variables $(d_j, d_k) \in \mathcal{S}$, the absolute difference between their correlations in the two matrices, $\text{Corr}_{R^P}^{(d_j, d_k)}$ and $\text{Corr}_{R^E}^{(d_j, d_k)}$, is defined as

$$\Delta_{jk}^{P,E} = |R_{jk}^P - R_{jk}^E|.$$

For the pair of matrices R^P and R^E , $\Delta_{jk}^{P,E}$ is considered the distance between the variables (d_j, d_k) .

Consequently, the mean distance between the variables $\forall (d_j, d_k) \in \mathcal{S}$ is defined as

$$\bar{\Delta}^{P,E} = \frac{1}{|\mathcal{S}|} \sum_{(j,k) \in \mathcal{S}} \Delta_{jk}^{P,E}.$$

That is, the average of the sum of all possible $\Delta_{jk}^{P,E}$.

Furthermore, if

$$\Delta^{P,E} = \{\Delta_{ih}^{P,E} : (d_i, d_h) \in \mathcal{S}\}$$

$p90(\Delta^{P,E})$ is the set of pairs $(d_i, d_h) \in \mathcal{S}$ for which $\Delta_{ih}^{P,E}$ lies at the 90th percentile of the set.

The total distance between correlation matrices P and E is then defined, with N_E denoting the sample size for imputation strategy E , as

$$C^{P,E} = \frac{\lambda \overline{\Delta}^{P,E} + (1 - \lambda) p_{90}(\Delta^{P,E})}{N_E}, \quad 0 \leq \lambda \leq 1. \quad (\text{score})$$

The parameter λ determines the weight assigned to the mean absolute difference, while $1 - \lambda$ determines the weight assigned to the 90th percentile.

Thus, the criterion does not depend solely on the average change in correlations, but also penalizes strategies that generate large differences for a relevant proportion of pairs of variables. Sample size is also taken into account, prioritizing strategies that retain a larger number of observations.

Consequently, an imputation strategy E is considered closer to the pairwise matrix than a strategy E' if

$$C^{P,E} < C^{P,E'}.$$

In this case, $\lambda = 0.5$ was used, assigning equal weight to the mean difference and the 90th percentile of the absolute differences.

C.3.3. Search for optimal hyperparameters

In MICE, it is possible to select how many samples to draw from the generated distribution function, then average them to obtain a more robust estimate of the missing value.

It is therefore reasonable to consider searching across different possible strategies for treating problematic cells. To this end, a set of candidate strategies $\mathcal{T}$ was defined, including alternatives such as MICE imputation, median imputation, mean imputation, and different combinations for genuine blanks and code 5 responses.

Since not all strategies depend on the same parameters, a single common hyperparameter space was not defined. Instead, each strategy $T \in \mathcal{T}$ was associated with its own configuration space $\mathcal{H}_T$. An element $h \in \mathcal{H}_T$ represents a specific configuration of that strategy. This configuration may include one or more decisions, such as the number of MICE imputations for genuine blanks, the number of MICE imputations for code 5 responses, or a tolerance criterion for the number of problematic cells allowed per observation.

In this sense, h should not necessarily be interpreted as a tuple with a fixed dimension. Its form depends on the strategy considered. For example, a strategy based only on the median may require no additional parameters, whereas a combined strategy may include several decisions, such as imputing genuine blanks with MICE, imputing code 5 responses with MICE, and filtering out observations with more than a certain number of problematic cells.

Thus, the full set of alternatives evaluated can be written as

$$\mathcal{A} = \bigcup_{T \in \mathcal{T}} (\{T\} \times \mathcal{H}_T).$$

Each element $a = (T, h) \in \mathcal{A}$ represents a specific alternative for treating the data. For each alternative a , a Spearman correlation matrix R^a was constructed and compared with the pairwise correlation matrix R^P , taken as the reference. The latter was used because it calculates each correlation using the observations available for that pair of variables, without imposing prior imputation on the entire matrix.

For reasons of computational complexity, the search was conducted over a finite set of configurations. For the parameters associated with MICE, values in $\{2, 5, 10, 15\}$ were considered. For strategies that depended on filtering criteria or tolerance of problematic cells, the corresponding values defined for each case were evaluated.

The general criterion was to choose the alternative that minimized the distance to the pairwise correlation matrix, according to the criterion defined in equation (score). In general terms, the selection can be expressed as

$$a^* = \arg \min_{a \in \mathcal{A}} C^{P,a}.$$

That is, the strategy and configuration that produced the correlation matrix closest to the pairwise matrix were selected, also taking into account the resulting sample size under that alternative.

Table C.3 shows the distance between the pairwise correlation matrix and some of the imputation methods tested.

Table C.3: Ranking of imputation strategies by distance to the pairwise Spearman correlation matrix.

Rank	Strategy	m_{blanks}	m_5	i	N_E	Mean	$p90$	score
1	MICE genuine blanks + MICE 5	2	2	–	18.977	0,0214	0,0440	$1,74 \times 10^{-6}$
2	MICE 5 + median genuine blanks $< i$	–	10	1	14.665	0,0171	0,0395	$1,93 \times 10^{-6}$
3	(MICE 5 + median genuine blanks) $< i$	–	1	6	12.358	0,0145	0,0349	$2,00 \times 10^{-6}$
4	MICE 5 + median genuine blanks $< i$	–	1	6	15.831	0,0208	0,0436	$2,03 \times 10^{-6}$
5	Complete-case	–	–	–	4.697	0,0478	0,0739	$1,30 \times 10^{-5}$

As shown in the table, the best alternative evaluated was MICE genuine blanks + MICE 5. Let us denote this strategy by $T_{\text{Blanks},5}^{\text{MICE}}$, where Blanks represents genuine blanks and 5 represents code 5 responses. For this particular strategy, the configuration space evaluated was

$$\mathcal{H}_{T_{\text{Blanks},5}^{\text{MICE}}} = \{(m_{\text{Blanks}}, m_5) : m_{\text{Blanks}}, m_5 \in \{2, 5, 10, 15\}\}.$$

Each configuration $h \in \mathcal{H}_{T_{\text{Blanks},5}^{\text{MICE}}}$ defines a specific alternative $a = (T_{\text{Blanks},5}^{\text{MICE}}, h)$. The comparison of the alternatives evaluated for the strategy “impute code 5 responses and genuine blanks using MICE” is presented below in Table C.4.

Table C.4: Grid search results for MICE genuine blanks $\rightarrow$ MICE 5.

m_{blanks}	m_5	Spearman score ($\times 10^{-6}$)
5	10	1,66
15	10	1,66
10	10	1,66
10	15	1,66
5	15	1,66
15	15	1,67
15	5	1,67
10	2	1,67
10	5	1,68
15	2	1,68
5	5	1,69
5	2	1,69
2	15	1,70
2	10	1,71
2	2	1,74
2	5	1,76

Although the numerical minimum of the grid search does not correspond exactly to the combination $(m_{\text{Blanks}}, m_5) = (2, 2)$, this strategy was adopted on the basis of a conservative methodological criterion. The difference in distance relative to the best combination observed is small in absolute terms (approximately 5%), while the computational cost rises rapidly when larger values of m_{Blanks} or m_5 are considered.

In particular, the combination $(2, 2)$ requires generating far fewer imputed datasets than alternatives such as $(5, 10)$, $(10, 10)$, or $(15, 15)$. Since the differences in the correlation matrix are small and the inferential conclusions do not change substantially, using a much more costly strategy solely to obtain a marginal improvement in the score $C^{P,a}$ does not seem justified.

Moreover, because this is a stochastic procedure, small differences in the ranking may depend on the seeds used. Therefore, when the grid search yields values that are very close to one another, it is not advisable to insist on choosing the minimum over the other options.

Structural blanks were treated differently from genuine blanks. These values do not correspond to nonresponse by respondents, but to questions that did not apply because of the questionnaire design, the route followed, or the characteristics of the branch. It is therefore inappropriate to impute them as if they were observed responses. For this reason, they were assigned a value of zero, which required centering all the variables.

In this way, the model does not interpret non-applicability as a positive, negative, or intermediate evaluation of the feature. The variable simply does not contribute to explaining overall satisfaction for that observation.

C.3.4. Structural blanks and centering variables

For the simple satisfaction model, both pooled and split by store format, structural blanks represent a different case from genuine missing values. These

do not represent an absence of response by the respondent, but situations in which the feature did not apply to the observation because the question was not part of the corresponding route through the questionnaire.

It is therefore inappropriate to impute a possible response in these cases. However, to include the feature in the modeling matrix, these values were assigned a value of 0 after centering the variable.

Let S_{u,d_j} be an indicator associated with observation u and feature d_j . We define $S_{u,d_j} = 1$ if feature d_j corresponds to a structural blank for observation u , and $S_{u,d_j} = 0$ if that feature was part of the corresponding route.

To calculate the center of the variable, only observations for which the feature was applicable and had a defined value are considered. Let

$$\mathcal{A}_j = \{u \in \mathcal{B}^{\text{final}} : S_{u,d_j} = 0, X_{u,d_j} \neq \text{NA}\}.$$

The mean of feature d_j is then defined as

$$\mu_j = \frac{1}{|\mathcal{A}_j|} \sum_{u \in \mathcal{A}_j} X_{u,d_j}.$$

Using this mean, the centered variable $\tilde{X}_{u,d_j}$ is constructed as follows:

$$\tilde{X}_{u,d_j} = \begin{cases} X_{u,d_j} - \mu_j, & S_{u,d_j} = 0, \\ 0, & S_{u,d_j} = 1. \end{cases}$$

Thus, when the feature was applicable, its value centered on the mean was used. When the feature did not apply because it was a structural blank, the value 0 was assigned.

Centering is important because it allows 0 to be interpreted as the absence of deviation from the feature's mean, rather than as a response category. Therefore, assigning 0 to a structural blank is not equivalent to imputing a response, but indicates that the variable contributes no differential effect for that observation.

Indeed, if feature d_j enters the model through the term

$$\beta_j \tilde{X}_{u,d_j},$$

then, for an observation with a structural blank in d_j , we have

$$S_{u,d_j} = 1 \quad \Rightarrow \quad \tilde{X}_{u,d_j} = 0 \quad \Rightarrow \quad \beta_j \tilde{X}_{u,d_j} = 0.$$

Therefore, the observation remains in the model, without being assigned an artificial response to a question it was not required to answer.

C.4. Heteroscedasticity diagnosis and HC3 robust standard errors

To justify the use of robust standard errors, we examined whether the homoscedasticity assumption was reasonable. This matters because, if the variance of the errors is not constant, conventional standard errors may provide an incorrect measure of coefficient uncertainty. In that case, the estimated coefficients do not change, but the way their uncertainty is calculated does. Thus, when there is evidence of heteroscedasticity, it is more appropriate to report robust standard errors, following the idea proposed by White [18] and the subsequent corrections by MacKinnon and White [19]. Since the model was estimated on different imputed datasets, the largest p -value observed across imputations is reported. In other words, the least favorable case for rejecting homoscedasticity is taken. The results are shown in Table C.5.

Table C.5: Heteroscedasticity tests on OLS residuals.

Test	Maximum p -value across imputations
Breusch-Pagan / Koenker	$1,18 \times 10^{-45}$
Classical Breusch-Pagan	$2,23 \times 10^{-55}$
White	$1,06 \times 10^{-152}$

As a visual complement, a scale-location plot was also constructed for Model 1 estimated on the first imputed dataset. This plot, Figure C.4, relates the model's fitted values to the square root of the absolute value of the standardized residuals. That is, the horizontal axis shows the value predicted by the model,

while the vertical axis shows $\sqrt{|r_i^*|}$, where r_i^* is the standardized residual of observation i .

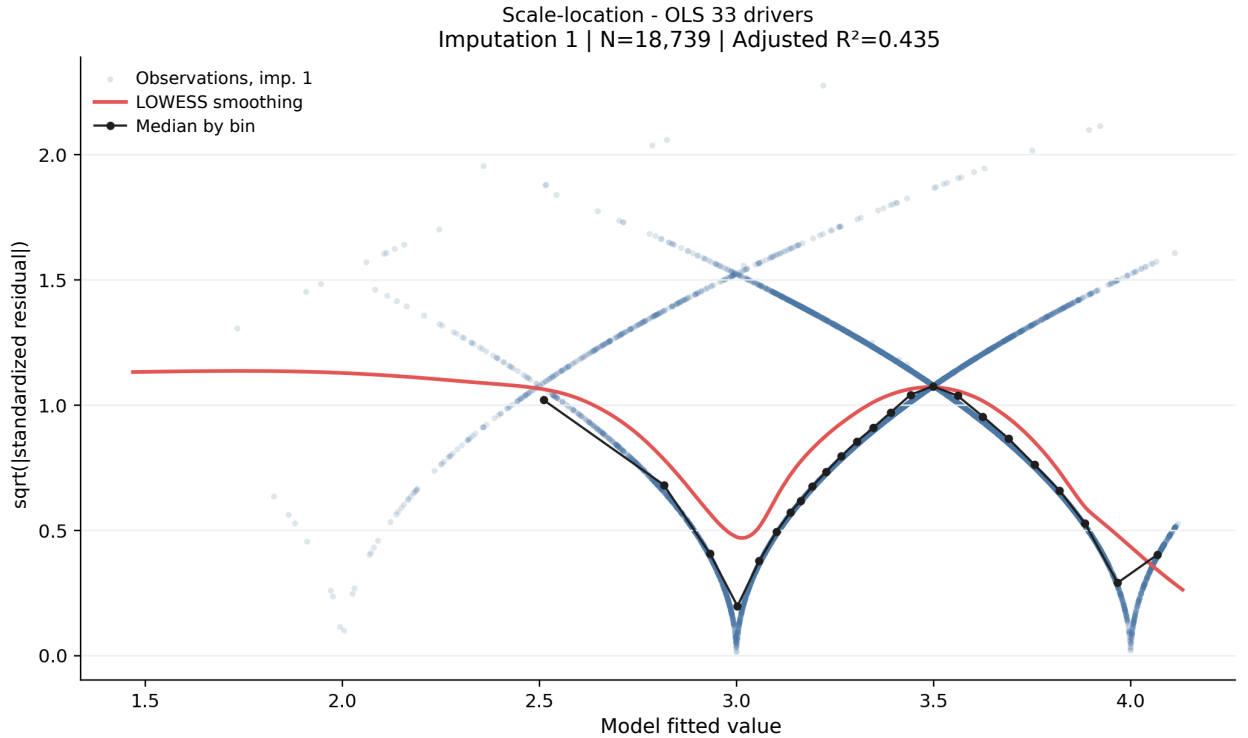


Figure C.4: Scale-location plot for Model 1 on the first imputed dataset.

This diagnostic allows us to observe whether the dispersion of the residuals remains approximately constant across fitted values. Under homoscedasticity, the smoothed line should be relatively horizontal, and the medians by interval should not show large changes. In contrast, the figure shows that dispersion is not completely stable across the range of predicted values. This is consistent with the formal tests reported in Table C.5.

Based on these results, the homoscedasticity assumption does not appear appropriate for this model. On the one hand, all three formal tests reject the null hypothesis with extremely low p -values. On the other, the scale-location plot shows that residual dispersion does not remain completely stable across predicted values.

This does not imply changing the coefficients estimated by OLS, but rather the way the uncertainty associated with those coefficients is calculated. For this reason, HC3 robust standard errors were reported. These standard errors allow inference without depending so heavily on the assumption of constant variance of the error term.

HC3 was chosen because it is a more conservative correction than other robust variants, such as HC1 or HC2. In particular, HC3 gives greater consideration to the possible presence of observations with high leverage, that is, observations whose combinations of explanatory-variable values lie farther from the general pattern of the sample. This may make sense in this case, given that the sample is large but also displays imbalance and clear evidence of heteroscedasticity.

C.5. Response-variable imbalance and weighted models

When the response variable is highly imbalanced, the ordinary least squares fit may be strongly influenced by the most frequent categories. The reason is that OLS gives equal weight to each observation, not to each category. Therefore, a category with many observations has a greater aggregate weight in the sum of errors that the model seeks to minimize

$$\min_{\beta} \sum_{i=1}^N (y_i - \hat{y}_i)^2.$$

To partially correct this problem, weighted models were estimated. In this case, each observation receives a weight w_i , so that errors in the less frequent categories have a greater influence on the fit

$$\min_{\beta} \sum_{i=1}^N w_i (y_i - \hat{y}_i)^2.$$

Let $g(i)$ be the category to which observation i belongs, and let $n_{g(i)}$ be the number of observations in the category of observation i . The weights were defined as

$$w_i = \frac{N}{G \cdot n_{g(i)}},$$

where N represents the total sample size and G the number of categories considered. Thus, less frequent categories receive a larger individual weight, while more frequent categories receive a smaller individual weight.

Tables C.6 and C.7 below show the weights calculated for the two weighting approaches considered. The first uses the four original categories of overall satisfaction, while the second combines some categories to reduce the extreme imbalance in the response variable.

Table C.6: Weights used for weighting with four categories of overall satisfaction.

Q5	Label	N	w_i
4	Very satisfied	7.703	0,608
3	Satisfied	10.210	0,459
2	Slightly satisfied	784	5,98
1	Dissatisfied	42	112

Table C.7: Weights used for weighting with aggregated categories of overall satisfaction.

Q5	Label	N	w_i
4	Very satisfied	7.703	0,811
3	Satisfied	10.210	0,612
1 2	Low satisfaction	826	7,56

Comparing the two tables reveals an important difference. When the four categories are weighted separately, the least frequent category receives a weight of 112. In contrast, grouping the lower-satisfaction categories reduces the weight of the low-satisfaction group to 7.56.

For this reason, the aggregated version is more stable. It retains the lower-satisfaction cases in the analysis while preventing a category with very few observations from carrying an excessively high weight in the estimation.

C.6. Evaluating temporal robustness across models

Temporal cross-validation (CV) was used to evaluate the models out of sample. The idea was to respect the order of the survey waves, so that the model is trained on past information and evaluated on later observations.

An expanding-window structure was used. In other words, in each fold the training set includes all the survey waves available up to a given point, and validation is performed on a subsequent time block. Thus, each validated observation is predicted by a model that did not use it during estimation.

This yields out-of-sample predictions, also known as OOF (out-of-fold) predictions. These predictions were used to calculate the out-of-sample metrics used to compare model performance: MAE, RMSE, and OOF R^2 .

To calculate the OOF RMSE, the predictions obtained in the validation blocks of the four folds were first concatenated. If V_f denotes the set of observations validated in fold f , the OOF set can be written as

$$V_{\text{OOF}} = \bigcup_{f=1}^F V_f,$$

where $F = 4$ and each observation belongs to the validation block of a single fold. Thus, all the predictions included in the calculation were generated by models that did not use those observations during estimation.

Using the concatenated predictions, the weighted OOF RMSE was calculated as

$$\text{RMSE}_{\text{OOF}} = \sqrt{\frac{\sum_{f=1}^F \sum_{i \in V_f} w_i \left(y_i - \hat{y}_i^{(-f)} \right)^2}{\sum_{f=1}^F \sum_{i \in V_f} w_i}},$$

where y_i is the observed value, $\hat{y}_i^{(-f)}$ is the prediction for observation i obtained with the model trained without using the validation block V_f , and w_i is the weight assigned to that observation.

For the WLS models, the OOF RMSE was calculated using the weights w_i . For the OLS model, the metric was calculated without weighting, which is equivalent to setting $w_i = 1$ for all observations. The only difference between the two cases is therefore the inclusion of the weights within the sums.

It is important to clarify that this calculation does not consist of averaging the RMSE values obtained in each fold. Because the validation blocks differ in size and may have different sums of weights, a simple average would give all folds equal importance. In contrast, concatenating the OOF predictions and calculating a single metric allows each observation to contribute according to its weight in the complete validation set.

The OOF R^2 was calculated following the same logic. The predictions from the validation blocks were first concatenated, and the model error was then compared with the total variability of the target variable within the OOF set. For the weighted models, the expression used was

$$R_{\text{OOF}}^2 = 1 - \frac{\sum_{f=1}^F \sum_{i \in V_f} w_i \left(y_i - \hat{y}_i^{(-f)} \right)^2}{\sum_{f=1}^F \sum_{i \in V_f} w_i \left(y_i - \bar{y}_{w, \text{OOF}} \right)^2},$$

where

$$\bar{y}_{w, \text{OOF}} = \frac{\sum_{f=1}^F \sum_{i \in V_f} w_i y_i}{\sum_{f=1}^F \sum_{i \in V_f} w_i}$$

is the weighted mean of the target variable in the OOF set.

For the WLS models, the OOF R^2 was calculated using the weights w_i and the weighted mean $\bar{y}_{w,\text{OOF}}$. For the OLS model, $w_i = 1$ was used for all observations, so the weighted mean is replaced by the arithmetic mean of the target variable in the OOF set.

As with RMSE, a single R^2 was calculated after concatenating all the OOF predictions, rather than averaging the values obtained in each fold.

In addition to these metrics, two measures associated with the fit of the full model are reported: the model's adjusted R^2 and the general adjusted R^2 . These measures are neither OOF measures nor final-test results; they are included as additional information for comparing the models.

In each fold, MICE imputation was performed after separating training and validation, using only the observations in the training set. Those criteria were then applied to both the training set and the validation block.

Thus, the validation data were not used to learn the relationships needed for imputation. This prevents information from the evaluated block from entering the training process. At this stage, an averaged version of the imputations within each fold was also used. In other words, CV used a single imputed dataset per fold. Consequently, four separate models were not estimated and then combined using Rubin's rules, as is done in the final fit.

The variables were centered following the same logic. The means used to center the features were calculated using only the training set of each fold and then also applied to the validation set. The validated block therefore contributes neither to imputation nor to the calculation of the centers used to transform the variables.

In addition to the temporal CV folds, a final test set of later observations was held out. This test was used as an additional out-of-sample check. For this final evaluation, the model was trained on the survey waves available through July 2025 and evaluated on subsequent waves, from September 2025 to March 2026.

Imputation and centering followed the same logic as in the earlier folds: they were estimated using only the final training set and then applied to the final test set.

The final-test metrics were calculated in the same way, but using only the observations held out for that set.

Table C.8: Structure of temporal CV and the final test.

Fold / set	Training	Validation / test	Evaluation N
Fold 1	Jun. 2021 – Dec. 2023	Jan. 2024 – May 2024	1.974
Fold 2	Jun. 2021 – May 2024	Jul. 2024 – Nov. 2024	1.495
Fold 3	Jun. 2021 – Nov. 2024	Jan. 2025 – Mar. 2025	568
Fold 4	Jun. 2021 – Mar. 2025	May 2025 – Jul. 2025	1.198
Final test	Jun. 2021 – Jul. 2025	Sep. 2025 – Mar. 2026	1.770

Table C.9 shows the out-of-sample metrics for the three models considered. The OOF metrics from temporal CV and the final-test metrics are reported. It should be emphasized that all metrics are weighted, except those for OLS 33 drivers.

Table C.9: Out-of-sample and final-test performance metrics.

Model	MAE OOF	RMSE OOF	R^2 OOF	MAE test	RMSE test	R^2 test
OLS 33 drivers	0,337	0,433	0,448	0,326	0,420	0,467
WLS 3cat 33 drivers	0,413	0,540	0,602	0,400	0,525	0,619
WLS 4cat 33 drivers	0,538	0,708	0,599	0,439	0,558	0,751

Table C.10 shows metrics for the full model. These metrics are neither OOF measures nor final-test results; however, they are reported to supplement the comparison between models.

Table C.10: Fit metrics for the full model.

Model	R^2 adj. model	R^2 adj. general
OLS 33 drivers	0,436	0,436
WLS 3cat 33 drivers	0,605	0,314
WLS 4cat 33 drivers	0,727	0,0683

As Table C.9 shows, the OLS model has the lowest overall out-of-sample errors. Among the weighted models, however, WLS 3cat has lower errors than WLS 4cat, both for the OOF metrics and for the final test.

Furthermore, as Table C.10 shows, WLS 4cat achieves the highest model adjusted R^2 , but has greater out-of-sample errors than WLS 3cat. In contrast, WLS 3cat corrects part of the imbalance without increasing prediction errors as much. This difference can also be seen in Table C.11, which compares the number of significant drivers, the SEs, and the confidence intervals for each model.

Table C.11: Precision of the estimated coefficients by model.

Model	Drivers $p < 0,05$	SE med.	SE Mean	95% CI med.	95% CI Mean	95% CI Max.
OLS 33 drivers	22	0,00890	0,00880	0,0347	0,0345	0,0449
WLS 3cat 33 drivers	16	0,0211	0,0214	0,0825	0,0837	0,108
WLS 4cat 33 drivers	4	0,0781	0,0790	0,306	0,310	0,535

C.7. Comparing feature-selection criteria and tuning SFS

C.7.1. Calculating the approximate contribution of each block in the PCA model

As a complementary analysis, an approximate contribution by block was calculated for the model estimated using the first three principal components. This calculation was used to interpret how the weight of the PCA model was distributed across the conceptual blocks of drivers.

The estimated model can be written as

$$Q5_i = \alpha + \beta_1 PC1_i + \beta_2 PC2_i + \beta_3 PC3_i + \text{controles}_i + \varepsilon_i.$$

In turn, each principal component is a linear combination of the original drivers. If x_{ij} denotes the value of driver j for observation i , and l_{jk} denotes the loading of driver j on component k , then

$$PCk_i = \sum_{j=1}^K l_{jk} x_{ij}.$$

Therefore, the approximate contribution of driver j through component k was calculated as

$$\text{aporte}_{j,k} = \beta_k l_{jk}.$$

Then, to move from drivers to blocks, the contributions of all drivers belonging to each block were added together. If B_b denotes the set of drivers in block b , the contribution of block b through component k was defined as

$$A_{b,k} = \sum_{j \in B_b} \beta_k l_{jk}.$$

Finally, the net contribution of the block was obtained by summing the contributions of the three components considered:

$$A_b^{\text{neto}} = A_{b,1} + A_{b,2} + A_{b,3}.$$

This calculation should be interpreted as an approximation. The PCA model does not estimate a direct coefficient for each driver, but rather for components that combine several drivers at the same time. Thus, the coefficient of each component is distributed among the drivers using their loadings, and these contributions are then summed within each block.

C.7.2. Comparing feature-selection criteria

After analyzing the model based on principal components, different feature-selection criteria were compared for the WLS 3cat model. The idea was to assess whether the number of drivers could be reduced without losing out-of-sample performance, while also ensuring that the model could be readily interpreted.

This comparison is important because the methods do not all approach the problem in the same way. The PCA model summarizes information in components, but its coefficients do not correspond directly to drivers from the questionnaire. In contrast, direct-selection methods retain the original variables. This makes it possible to interpret the model in terms of specific drivers of the in-store experience.

All criteria were compared using the same imputation, centering, and WLS 3cat weighting strategy. Thus, the observed differences are attributable to the criterion used to select features, rather than to changes in the construction of the modeling matrix.

For SFS, selection was performed sequentially, using the minimization of mean squared error as the internal criterion. In other words, at each step the driver that produced the greatest improvement under that criterion was added. Then, to compare SFS with the other methods, the resulting models were evaluated using the same out-of-sample metrics: weighted OOF MAE, RMSE, and R^2 .

Therefore, Table C.12 does not necessarily compare the internal criterion used by each method, but rather the final performance of the models obtained under each selection strategy. This makes it possible to assess whether reducing the number of drivers improves or maintains performance relative to the full model.

Table C.12: Comparison of feature-selection criteria for the WLS 3cat model.

Method / specification	k drivers	MAE OOF	RMSE OOF	R^2 OOF
Weighted WLS 3cat	33	0,413	0,540	0,602
Weighted SFS 3cat	20	0,412	0,538	0,605
Weighted Permutation 3cat	28	0,413	0,539	0,604
Weighted ElasticNetCV 3cat	27	0,413	0,539	0,603
Weighted RFE 3cat	17	0,412	0,539	0,603
Robust bootstrap WLS 3cat	8	0,414	0,542	0,599

As shown, the weighted SFS 3cat model has the best OOF RMSE and R^2 . It also essentially ties with RFE on MAE. Although RFE achieves a slightly lower MAE, the difference is minimal, whereas SFS has a lower out-of-sample RMSE and a higher out-of-sample R^2 . For this reason, SFS was prioritized as the main selection criterion.

The Permutation and ElasticNetCV methods also select a large number of drivers, close to the full model's 33 variables, without improving out-of-sample performance. In this respect, they do not appear to provide a sufficient reduction in model complexity. The robust bootstrap criterion, in contrast, selects a much smaller model with 8 drivers, but with a greater deterioration in the evaluation metrics.

SFS therefore offers a reasonable middle ground. It reduces the number of drivers, preserves a direct interpretation of the selected variables, and does not cause a significant loss of out-of-sample performance compared with the full model. For this reason, it was used as the main criterion for defining the reduced model.

C.7.3. Tuning the SFS auto-k criterion

Once SFS had been chosen as the main selection criterion, the number of drivers to include in the reduced model still had to be defined. An auto-k version was used instead of setting a value of k manually. The idea was to evaluate different tolerance values and let the procedure select the number of drivers based on the marginal improvement observed.

In this study, this improvement was defined as the reduction in weighted MSE under the weighted 3cat metric in temporal validation when one more driver was added to the model. That is, if MSE_k denotes the error of the model with k drivers, the improvement from $k - 1$ to k drivers can be written as

$$\Delta MSE_k = MSE_{k-1} - MSE_k.$$

On this basis, the procedure continues adding drivers as long as the reduction in error is greater than the specified tolerance:

$$MSE_{k-1} - MSE_k > \text{tol.}$$

When this improvement no longer exceeds the tolerance, the procedure stops. This may make sense if one considers that adding drivers generally reduces MSE substantially at first, but eventually each additional driver contributes less and less. At that point, continuing to add variables increases model complexity without meaningfully improving out-of-sample performance.

For each tolerance value in

$$tol = \{1 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 5 \times 10^{-3}, 1 \times 10^{-2}\}$$

a set of drivers selected by SFS was obtained, and the out-of-sample performance of the resulting model was then evaluated.

Table C.13: Grid search for the SFS auto-k criterion for the WLS 3cat model.

tol	Consensus k	MAE	MSE	RMSE	R^2
5×10^{-5}	20	0,410	0,290	0,538	0,601
1×10^{-5}	20	0,410	0,290	0,538	0,601
1×10^{-4}	20	0,410	0,290	0,538	0,601
1×10^{-3}	12	0,413	0,293	0,541	0,597
5×10^{-4}	15	0,413	0,293	0,541	0,597
5×10^{-3}	7	0,416	0,296	0,544	0,592
1×10^{-2}	5	0,423	0,308	0,555	0,575

As Table C.13 shows, tolerance values of 5×10^{-5} , 1×10^{-5} , and 1×10^{-4} produce the same result: they select 20 drivers and achieve the same out-of-sample metrics. Consequently, the grid-search optimum was associated with a model with $k = 20$ drivers and $tol = 10^{-4}$.

As the tolerance increases, the procedure becomes more restrictive and selects fewer drivers. This reduces model complexity, but also worsens the out-of-sample metrics. For example, with $tol = 0.001$, the model has 12 drivers and MSE rises to 0.293, whereas with $tol = 0.01$, the model has only 5 drivers and MSE rises to 0.308.

For this reason, SFS auto-k with $k = 20$ was chosen for modeling. This alternative maintains the best performance observed under the selection criterion while reducing the number of drivers relative to the full model with 33 variables.

C.8. Robustness of the loss aversion model

This section presents additional results from the loss aversion model. The aim is to complement the interpretation in the main section.

As mentioned earlier, the asymmetry measure was calculated for each driver as

$$\Delta_k = |\beta_k^-| - \beta_k^+.$$

This measure compares the magnitude of the effect associated with a low rating with the effect associated with a high rating. When $\Delta_k > 0$, the low rating carries greater weight, in absolute terms, than the high rating.

In the expected case, where $\beta_k^- < 0$ and $\beta_k^+ > 0$, this measure can be written as

$$\Delta_k = -\beta_k^- - \beta_k^+.$$

The symmetry test can therefore be formulated using the following null hypothesis:

$$H_0 : \beta_k^- + \beta_k^+ = 0,$$

$$H_1 : \beta_k^- + \beta_k^+ \neq 0.$$

In the expected case, this test is equivalent to testing

$$H_0 : \Delta_k = 0.$$

Using the estimated coefficients, the asymmetry measure is calculated as

$$\hat{\Delta}_k = -\hat{\beta}_k^- - \hat{\beta}_k^+.$$

Because both coefficients are estimated within the same model, the standard error of $\hat{\Delta}_k$ must account not only for the variance of each estimator, but also for the covariance between them. Therefore,

$$SE(\hat{\Delta}_k) = \sqrt{\text{Var}(\hat{\beta}_k^-) + \text{Var}(\hat{\beta}_k^+) + 2 \text{Cov}(\hat{\beta}_k^-, \hat{\beta}_k^+)}.$$

The variances and covariance are obtained from the model's variance-covariance matrix. This accounts for the fact that the two coefficients were not estimated independently.

The statistic used for the test is

$$z_k = \frac{\hat{\Delta}_k}{SE(\hat{\Delta}_k)}.$$

At a significance level of 5%, H_0 is rejected when

$$|z_k| > 1.96.$$

Equivalently, the confidence interval can also be constructed as

$$IC_{95\%}(\Delta_k) = \hat{\Delta}_k \pm 1.96 SE(\hat{\Delta}_k).$$

If this interval does not contain zero, the difference between the two magnitudes is considered statistically significant.

If the null hypothesis is rejected, there is evidence that the two effects are not symmetric for that driver. However, the direction of the asymmetry depends on the sign of $\hat{\Delta}_k$. When $\hat{\Delta}_k > 0$, the asymmetry is in the direction expected under loss aversion, since the penalty associated with a low rating is greater than the reward associated with a high rating. In contrast, when $\hat{\Delta}_k < 0$, the reverse occurs: the positive effect is larger in magnitude than the negative effect, which can be interpreted as reward dominance.

It is important to clarify that this test is not the same as examining whether each coefficient is significant separately. The main model evaluates whether the coefficient associated with a low rating differs from zero and whether the coefficient associated with a high rating also differs from zero, both relative to the reference category. In contrast, the asymmetry test evaluates whether the difference between the two magnitudes is statistically different from zero.

This makes sense if one considers that the model could show that a low rating subtracts and a high rating adds, while the difference between the two effects might not be large enough to establish that the penalty is greater than the

reward. Therefore, to speak of loss aversion in statistical terms, it is not enough for $\hat{\Delta}_k > 0$: that difference must also be significant.

As an additional check, the full model with 33 drivers was compared with a reduced version obtained by applying SFS auto-k to the loss aversion model. The idea was to check whether reducing the number of variables substantially changed the positive coefficients, negative coefficients, and asymmetry measures.

For this comparison, SFS auto-k was applied at the driver level, rather than to individual dummy variables. That is, each driver entered or left the model with both of its indicators: low rating (1||2) and high rating (4). This preserves the comparison between the two effects.

Table C.14: Comparison between the full model and the reduced SFS auto-k model for loss aversion.

Metric	Value
Adjusted R^2 , full model with 33 drivers	0,622
Adjusted R^2 , SFS auto-k loss aversion model	0,609
Correlation of positive coefficients	0,985
Correlation of negative coefficients	0,978
Correlation of asymmetry measures	0,950

The correlations in Table C.14 were calculated for the $K = 15$ drivers selected by the SFS auto-k procedure. In other words, the coefficients estimated for those same drivers in the full model with 33 drivers and in the reduced model were compared.

As shown, the reduced model has a somewhat lower adjusted R^2 than the full model, but retains high correlations for the positive coefficients, negative coefficients, and asymmetry measures. This suggests that reducing the number of variables does not substantially change the overall interpretation of the loss aversion model.

Appendix E

LIBRARIES USED IN THE COMPUTATIONAL IMPLEMENTATION

This appendix presents the main libraries used in the implementation. The aim is not to describe in detail how each one works, but to document the tools used for data processing, variable construction, imputation, model estimation, and the generation of results.

Table E.1: Main libraries used in the analysis.

Library	Main use
pandas	Reading, cleaning, transforming, and joining datasets; preparing tables and exporting results.
numpy	Numerical computation, array operations, masks, means, errors, and matrix operations.
matplotlib	Generating the main plots, such as bar charts, heatmaps, coefficient plots, curves, and diagnostics.
openpyxl	Writing and formatting Excel files, including worksheets, styles, colors, tables, and image insertion.
statsmodels	Estimating statistical and econometric models, such as OLS and WLS, formula-based models, diagnostics, tests, and inference.
scipy	Statistical tests, correlations, hierarchical clustering, distances, and distribution functions.
scikit-learn	MICE imputation, PCA, standardization, cross-validation, variable selection, and predictive metrics.
Pillow / PIL	Targeted editing of images and figures, such as adding legends or visual references.
geopandas	Producing maps and geographic visualizations of stores.

As shown, the libraries serve different purposes within the study. Some were used mainly to prepare the dataset, others to estimate models and perform statistical tests, and others to generate plots or supplementary outputs.

Bibliography

- [1] W. Bruce Traill. The rapid rise of supermarkets? nature, causes, and impacts. *Development Policy Review*, 24(2):163–174, 2006.
- [2] Nic S. Terblanche. Revisiting the supermarket in-store shopping experience. *Journal of Retailing and Consumer Services*, 40:48–59, 2018.
- [3] Óscar González-Benito. Spatial and structural analysis of retail competition. *International Journal of Retail & Distribution Management*, 33(3):218–233, 2005.
- [4] Dhruv Grewal and Michael Levy. Retailing research: Past, present and future. *Journal of Retailing*, 83(4):447–452, 2007.
- [5] Omid Kamran-Disfani, Murali K. Mantrala, Alicia Izquierdo-Yusta, and María Pilar Martínez-Ruiz. The competitive impacts of a new store format: A structural analysis. *Journal of Retailing*, 93(4):453–469, 2017.
- [6] Katri Koistinen and Raija Järvinen. Consumer grocery shopping profiles: Connections between shopping behavior and store format choice. *International Journal of Retail & Distribution Management*, 37(1):54–77, 2009.
- [7] Michael Levy, Barton A. Weitz, and Dhruv Grewal. *Retailing Management*. McGraw-Hill/Irwin, Boston, 8th edition, 2010.
- [8] Thomas Reutterer and Christoph Teller. Store format choice and shopping trip types. *Journal of Business Research*, 62(12):1199–1210, 2009.
- [9] Patrick Van Kenhove, Walter Van Waterschoot, and Kristof De Wulf. Impact of task definition on store-attribute saliences and store choice. *Journal of Retailing*, 75(1):125–137, 1999.
- [10] Pratibha A. Dabholkar, Dayle I. Thorpe, and Joseph O. Rentz. A measure of service quality for retail stores: Scale development and validation. *Journal of the Academy of Marketing Science*, 24(1):3–16, 1996.
- [11] Marcel Goić, Camilo Levenier, and Ricardo Montoya. Drivers of customer satisfaction in the grocery retail industry: A longitudinal analysis across store formats. *Journal of Retailing and Consumer Services*, 60:102505, 2021.
- [12] Eugene W. Anderson and Vikas Mittal. Strengthening the satisfaction-profit chain: The asymmetric effects of negative and positive performance. *Journal of Marketing Research*, 37(4):421–434, 2000.
- [13] Daniel Kahneman and Amos Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979.
- [14] Shin’ya Suzuki, Reginald B. Mitchell, and Rajshekhar G. Javalgi. Asymmetric effects of service quality on customer satisfaction in international services. *Journal of Services Marketing*, 15(7):542–558, 2001.

- [15] Amos Tversky and Daniel Kahneman. Loss aversion in riskless choice: A reference-dependent model. *The Quarterly Journal of Economics*, 106(4):1039–1061, 1991.
- [16] David R. Roberts, Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-Arroita, Severin Hauenstein, José J. Lahoz-Monfort, Boris Schröder, Wilfried Thuiller, David I. Warton, Brendan A. Wintle, Florian Hartig, and Carsten F. Dormann. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017.
- [17] Trivellore E. Raghunathan, James M. Lepkowski, John Van Hoewyk, and Peter Solenberger. A multivariate technique for multiply imputing missing values using a sequence of regression models. *Survey Methodology*, 27(1):85–95, 2001. Catalogue No. 12-001.
- [18] Halbert White. A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4):817–838, 1980.
- [19] James G. MacKinnon and Halbert White. Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics*, 29(3):305–325, 1985.